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Refining footprint analyses beyond the national level: An open tracing framework to capture the heterogeneity of subnational commodity flows

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Affidavit

I hereby declare that I have authored this master thesis independently, and that I have not used any assistance other than that which is permitted. The work contained herein is my own except where explicitly stated otherwise. All ideas taken in wording or in basic content from unpublished sources or from published literature are duly identified and cited, and the precise references included.

I further declare that this master thesis has not been submitted, in whole or in part, in the same or a similar form, to any other educational institution as part of the requirements for an academic degree.

I hereby confirm that I am familiar with the standards of Scientific Integrity and with the guidelines of Good Scientific Practice, and that this work fully complies with these standards and guidelines.

Vienna, 22.07.2022

Stefan TRSEK (*manu propria*)

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Abstract

Agricultural commodities are embedded in complex international supply chains, creating highly localized environmental pressures driven by global consumption. Adequately tracing commodity flows from local production to final consumption is thus key for assessing the spatial heterogeneity in resource footprints. However, existing approaches to refine macro-level footprints beyond the national level either neglect subnational flows by just proportionally downscaling national flows or rely on costly microdata and lack transparency and reproducibility. This thesis introduces a transparent and reproducible modeling framework to trace commodity flows from subnational production regions to importing countries, using the prominent case study of Brazilian soybeans. It demonstrates how openly available economic, environmental and geospatial data at different scales can be integrated to build fine-scale supply and use accounts, model subnational transport flows, and link them to international trade. The resulting commodity flows are benchmarked against data of the Trase initiative, revealing that the framework can improve the accuracy of footprint analyses compared to the commonly used proportional downscaling of footprints. The influence of model design and exogenous parameters on outcomes is assessed, and limitations of the approach are discussed. Notably, the framework is designed to be fully compatible with the Food and Agriculture Biomass Input-Output model FABIO, allowing the tracing of flows and associated environmental pressures from the point of production to downstream final consumption. With the advent of widely accessible spatial environmental data, the framework aims to provide a blueprint for other agricultural products and regions, and for the linkage of material flows to fine-scale environmental pressures and impacts.

Kurzfassung

Landwirtschaftliche Güter sind in komplexe internationale Lieferketten eingebettet, in denen lokale Umweltauswirkungen durch den weltweiten Endverbrauch angetrieben werden. Die adäquate Nachverfolgung von Güterströmen vom Produktionsort bis zum Punkt des Endkonsums ist deshalb essenziell, um die räumliche Heterogenität von Ressourcenfußabdrücken zu erfassen. Bestehende Ansätze zur Erhöhung der räumlichen Auflösung von Fußabdrücken beruhen jedoch entweder auf grobem proportionalem „Downscaling“ von der nationalen Ebene oder auf kostspieligen Mikrodaten mit Einschränkungen hinsichtlich Transparenz und Reproduzierbarkeit. Anhand des Fallbeispiels brasilianischen Sojas entwickelt diese Masterarbeit einen transparenten und reproduzierbaren Modellrahmen zur Verfolgung landwirtschaftlicher Güterflüsse von subnationalen Anbauregionen bis zu den Erstimportländern. Dabei wird gezeigt, wie frei verfügbare Wirtschafts-, Umwelt- und Geodaten auf verschiedenen Skalen integriert werden können, um räumlich detaillierte Aufkommens- und Verwendungstabellen zu erstellen, subnationale Transportströme zu modellieren, und diese mit internationalen Handelsströmen zu verknüpfen. Die resultierenden Güterströme werden mit Daten der Trase Initiative verglichen, wobei deutlich wird, dass das Modell eine erhöhte Genauigkeit gegenüber dem häufig praktizierten proportionalen Downscaling liefern kann. Zudem werden der Einfluss des Modelldesigns und exogener Parameter auf die Ergebnisse analysiert sowie Limitationen des Ansatzes diskutiert. Der Modellrahmen ist vollständig mit dem FABIO Input-Output-Modell kompatibel, wodurch die weitere Verfolgung der Güterströme entlang globaler Lieferketten bis zum Punkt des Endverbrauchs ermöglicht wird. Mit dem vermehrten Aufkommen offen zugänglicher räumlicher Umweltdaten soll das Modell als Blaupause für andere landwirtschaftlichen Produkte und Regionen sowie für die Verknüpfung von Materialflüssen mit lokalen Umweltbelastungen und -auswirkungen dienen.

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1 Introduction

Agricultural commodities are the core foundation of the food system and many non-food industries. Along their journey from primary production to final consumers, they are embedded in increasingly complex webs of international supply chains (Dellink et al., 2020, Humphrey and Memedovic, 2006). Consequently, local environmental impacts of agriculture are often driven by final consumption in distant places around the globe – a phenomenon also referred to as "telecouplings" (Kapsar et al., 2019, Liu et al., 2013). The crucial role of such distant socio-economic interactions in many critical environmental issues, such as deforestation and biodiversity loss, has provoked calls for increased supply chain tractability to reconcile the opaque links between production and consumption on a global scale (Lambin et al., 2018, McNevin et al., 2020).

However, many environmental pressures and impacts associated with agriculture are highly localized, and tracing methods have long dealt with a trade-off between coverage and resolution (Godar et al., 2016, Moran et al., 2020). On the one hand, macro-level footprinting and material flow analysis tools, often based on environmentally extended multi-regional input-output (MRIO) models (Minx et al., 2009, Wiedmann et al., 2006) or bilateral trade matrices (Kastner et al., 2011), can capture global commodity chains as a whole but typically rely on economic transaction data on the level of aggregate nations. Thus, they lack the resolution to adequately capture spatially heterogeneous environmental effects at local scales. Attempts to increase the spatial resolution of such models have mostly relied on simple proportional downscaling of national-level effects to the local level by using fine-scale environmental extensions or production data (Hoang and Kanemoto, 2021, Kanemoto et al., 2016, Lutter et al., 2016, Moran and Kanemoto, 2017, Sun et al., 2020). However, such approaches do not capture the heterogeneity of subnational flows and hence provide only a rough approximation of potential impact locations.

On the other hand, highly specific and detailed tools such as life-cycle analysis (Hauschild et al., 2018) provide fine-grained information on localized impacts and supply-chain linkages but are typically focused on a narrow set of representative samples of companies or products. Therefore, they lack the coverage to depict sector-wide flows on a large scale and assess related impacts across entire countries. On the micro-level, several innovative tracing technologies used by supply chain actors are promising (Badia-Melis et al., 2015), but they are not yet mature and established enough to derive comprehensive footprint assessments.

More recently, however, with the increasing availability of economic and environmental data at finer scales, a middle ground approach has emerged, aiming to bridge the gap between macro-level international and subnational flows. Prominently, the Transparency for Sustainable Economies (Trase) initiative (Trase, 2022), based on a framework initially proposed by Godar et al. (2015), models supply chains of several deforestation-risk commodities from high-resolution administrative areas (e.g., municipalities, departments) to countries of first import. In its current version, it makes use of customs records and company-specific asset data to link local production regions to shipments of individual trading companies (Trase, 2018a). Similar subnational model approaches have been put forward by Smith et al. (2017) for the case of the US corn complex or by Courtonne et al. (2015) for the French cereal supply chain. These approaches provide a major contribution towards more accurate footprint analyses, but methods so far lack reproducibility and generalizability. For example, while the supply chain results of the Trase platform are openly available, the method relies on costly customs and tax registration data and the source code is not made public. Other

contributions have so far focused on single case studies of countries in the Global North, which are less affected by globally telecoupled environmental impacts. As a result, there is little open infrastructure to replicate results and adapt models to new commodities or countries facing pressing environmental challenges driven by global trade.

This thesis introduces an open subnational supply chain modeling framework, applied to the prominent case of Brazilian soy. The framework is entirely based on freely available data and fully compatible with the Food and Agricultural Biomass Input-Output model FABIO to pave the way for replicable consumption-based footprint analyses at a high spatial resolution. To evaluate the model's performance, the resulting soy flows from the municipality of production to the countries of first import are benchmarked against the results of state-of-the-art methods, i.e. proportional downscaling and the Trase approach. This provides insights into the capability and reliability of the presented model. Moreover, two alternative transport optimization models to allocate subnational flows – a simple Euclidean distance minimization and a more complex network-based setting incorporating multiple transport modes – are tested. This aims to shed light on the added value of more realistic but complex model designs vis-a-vis easily implementable approximations. Finally, the thesis also addresses sources of uncertainty and limitations of such open-data modeling approaches. Through its transparency and openness, it aims to facilitate future extensions and improvements of the model.

The thesis is structured as follows: Section 2 provides some background information on the chosen case study of Brazilian soy. This case study is then used in section 3 to introduce the supply chain modeling framework, consisting of three main steps. The first step (3.1) establishes a subnational data foundation of all relevant supply and use items, based on a diverse range of open data. Step two (3.2) then models the spatially explicit commodity flows by means of the two alternative transport optimization settings mentioned above. In the third step (3.3), the subnational flows are connected to international export flows. Section 4 conducts a comparison of results with Trase data and a pure proportional downscaling. The discussion in section 5 highlights key implications, possible extensions and limitations of the framework before section 6 adds some concluding remarks. The data and R code scripts of all working steps are openly available via GitHub (<https://github.com/strsek/soyprint>).¹

2 The Brazilian soy complex: A textbook telecoupling example

The story of soybean cultivation in Brazil, its drivers and consequences can be seen as a remarkable example of the globalization of agriculture. Soybeans are not indigenous to South America, but over the last century, their rise has shaped the continent's agricultural landscape like no other crop. The soybean, a leguminous plant rich in proteins and fats, is autochthonous to East Asia and did not reach South America before the late 19th century (Hymowitz, 2008). Goldsmith (2008b) notes that commercial cultivation in Brazil only began in the 1960s in the south of the country but significantly expanded to the west and north after the development of varieties adapted to tropical climates in the 1970s. This evoked enormous potential for the crop, especially in the savannas of the Cerrado region.

¹Note that the Github repository also contains code for further working steps not covered by this thesis, namely the linkage of results with the input-output system of FABIO, the calculation of consumption-based footprints and the refinement of footprints to the grid level using land-use raster data.

In combination with comparably low operational costs, improving technology, targeted agricultural policies, and increasing demand, the Brazilian soybean complex has since experienced an unprecedented expansion (Goldsmith, 2008b, Martinelli et al., 2017, Yao et al., 2018). Figure 1 depicts the growth in soybean production and exports over the last six decades. With a production volume of 122 Mt in 2020, Brazil is now the world's largest soybean producer, accounting for roughly one third of global production. Only the United States, which was surpassed by Brazil in 2019, reaches similar production levels (FAOSTAT, 2022d).² In contrast to the US, however, the majority of Brazilian soy is exported (roughly two thirds). About half of the globally traded soybeans – one of the world's most traded agricultural commodities – thus stem from Brazil, not even including the trade of processed soybean cake and oil. This highlights the important role of foreign demand in the rise of Brazil's soy complex. Notably, soy has become the main item in the Brazilian trade balance (Comex Stat, 2022b).³

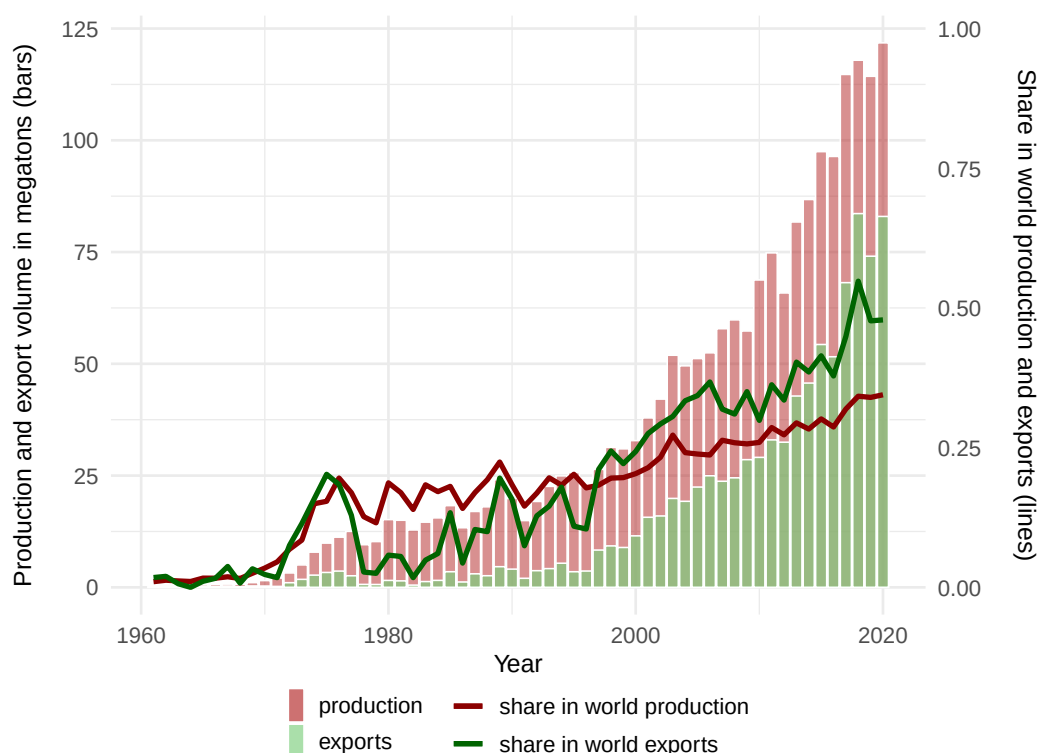


Figure 1: Trends in Brazilian soybean production and exports. Bars (left axis) depict absolute production and export volumes, and lines (right axis) indicate the respective shares in world production and exports. *Source:* Own visualization based on FAOSTAT (2022d) and FAOSTAT (2022e).

²The US recorded a production volume of 113 Mt in 2020. Within South America, Brazil accounted for 65% of production, with the remainder mainly covered by Argentina (49 Mt, the world's third-largest producer) and Paraguay (11 Mt).

³When combining soybeans, soybean cake and soybean oil, they accounted for a trade surplus of \$47.5 billion in 2021 (Comex Stat, 2022b). This even exceeds the surplus generated by iron ores (44.6) or petroleum oils (26.6). Other major agricultural products, such as sugar (9.2), poultry (6.9), beef (6.8), coffee (5.8) or maize (3.2), do not reach this level.

The rising demand for Brazilian soy was mainly fueled by increasing meat consumption, both domestically and globally (Kuepper et al., 2019, Yao et al., 2018). Due to its high protein content, soy – mainly in the form of soybean cake – is a key ingredient and the most important protein source in global livestock feeds (Stein et al., 2008). Considerable amounts of Brazil’s soy are used in domestic livestock industries (which are among the largest in the world and themselves produce high amounts for export). But the majority is exported – partly as processed cake, partly as raw soybeans – to eventually end up in foreign livestock sectors (Kuepper et al., 2019).⁴ This livestock-driven external demand was the main engine of the rapid export growth since the mid-90s (see figure 1), with China and the EU as main importers (Comex Stat, 2022b). The second (although less important) driver was the demand for soybean oil, which is an important edible oil both in Brazil and in major export destinations such as China (Belik, 2020, Nguema, 2019). More recently, it has also been increasingly used as a feedstock for biodiesel production, an industry fostered by the Brazilian government since the early 2000s (Lima et al., 2011).⁵ Thus, with the expansion of the primary soybean sector in Brazil, also the downstream processing industries (crushing and oil refining) experienced significant growth (Goldsmith, 2008b).

The rapid rise of the crop’s production has both been extensive, via the expansion of the soybean frontier (e.g., Arvor et al., 2013, Rausch et al., 2019, Song et al., 2021), and intensive, including the adoption of genetically modified variants of the plant (da Silva, 2019). Since the mid-1970s, the total soy cultivation area in Brazil increased by a factor of seven and now amounts to about 370 000 km², or roughly the size of Germany (IBGE, 2022a).⁶ With the partly unregulated expansion of the frontier deeper into the country, soy has played a significant role in land-use change (LUC) and the loss of native vegetation, most notably in the Cerrado and Amazon biomes (Song et al., 2021, Trase, 2018b, Walter and Garnett, 2020). Evidence suggests that much of the LUC related to soy cannot be directly attributed to the crop’s cultivation but is rather the outcome of indirect spatio-temporal processes, in which soy often replaces cattle pastures, which are then pushed further into the native vegetation (Arima et al., 2011, Barona et al., 2010, Kuschnig et al., 2021, Song et al., 2021). Concerns about deforestation have led to initiatives such as the soy moratorium, a sectoral agreement in which traders committed not to buy any soy from Amazon areas that were deforested after 2008 (Heilmayr et al., 2020). However, most of the soy-related land conversion and deforestation has affected the less protected Cerrado biome, particularly the MATOPIBA region covering the states of Maranhão, Tocantins, Piauí, and Bahia (Escobar et al., 2020, Polizel et al., 2021, Soterroni et al., 2019).

The high importance of export demand, triggered by the downstream demand for diverse food and energy products in major consumption regions, and the significant local impacts of Brazil’s soybean complex, made it one of the most prominent examples of the telecoupling concept (Kapsar et al., 2019, Liu et al., 2013). The significance of this concept for this particular supply chain is not only based on its high trade involvement, but also on the fact that its demand is mainly driven *indirectly* by the growing consumption of downstream products that embody soy as an intermediate input at some stage along their supply chain. Recent studies on the soy complex have increasingly focused on these distant socio-economic inter-

⁴Notably, only 2% of soybean protein is consumed directly by humans (e.g., as tofu, soy milk, etc.), with the remainder being almost entirely consumed by livestock (Goldsmith, 2008a).

⁵For more details on the domestic use structure of soybeans and their derivatives, see section 3.1.1.

⁶According to the IBGE’s Agricultural Survey, the harvested soy area was 5143 km² (14% of all temporary crops) in 1974 and 371 881 km² (48%) in 2020. The surface area of Germany is 357 137 km² (United Nations, 2013).

linkages and their environmental consequences, including land and water use (Kastner et al., 2011, Taherzadeh and Caro, 2019), LUC and deforestation (Lima et al., 2011, Millington et al., 2021, Pendrill et al., 2019, da Silva et al., 2021), carbon flows (Bharucha, 2014, Yao et al., 2018, Zaks et al., 2009) as well as socio-economic implications (Lima et al., 2011, Martinelli et al., 2017, da Silva et al., 2020, 2021). However, in the light of the country's magnitude and heterogeneity, the linkage of local environmental processes and their distant drivers has been limited by the absence of comprehensive subnational supply chain data, forcing researchers to rely either on average national effects or micro-level case studies. The subnational supply chain model introduced by Godar et al. (2015) and resulting in the Trase database was the first comprehensive framework to shed light on these telecoupled processes on a subnational level. This gave rise to a number of spatially detailed contributions based on the narrative of Brazilian soy, mostly in the context of land use, deforestation and emissions (zu Ermgassen et al., 2020, Escobar et al., 2020, Godar and Gardner, 2019, Green et al., 2019). Croft et al. (2018) used the Trase data in combination with an MRIO model to link the soy trade flows and land use further downstream to the consumption of final food products around the world.

The case of Brazilian soy can thus be seen as a catalyst in the advancement of the telecoupling research agenda, inspiring numerous methodological approaches that have also been applied to other agricultural products and primary resources (Friis and Nielsen, 2019, Kapsar et al., 2019). Based on its prominence and favorable data availability, I also make use of this particular crop and country to develop and demonstrate a spatially explicit open-source supply chain modeling framework. The refined perspective on commodity flows provided by the model can give a clearer and more accurate picture of the associated risks. As further discussed in section 5, this can inform both supply chain actors and policymakers to take more targeted actions towards more sustainable supply chains.

3 Methodology

This section describes the method and data sources of the model framework, demonstrated using the example of Brazilian soybeans. The workflow consists of three main steps: First, a subnational supply and use data inventory on the level of Brazil's municipalities is constructed as the basis of the model. Second, the subnational flows from local supply to local use are modeled by means of a transport optimization model. Third, the resulting subnational flows are linked with export data to trace flows further to the countries of first import.

All steps rely exclusively on openly available data and integrate a wide range of statistical and spatial information on different scales. The framework is implemented in R and GAMS and is fully reproducible. As the model is built to be fully compatible with FABIO, the year of analysis for the given case study is 2013, currently the most recent available year of the database.⁷

⁷Updates of FABIO for more recent years will be published in the near future.

3.1 Completing the subnational data basis

3.1.1 Data requirements

To comprehensively model the supply chain of a given commodity, information on both its supply and use is required as a starting point. On the supply side, goods can either be produced domestically or sourced from elsewhere. On the use side, they can be used domestically, either as intermediate inputs in other sectors or as final consumption goods, or they can be exported to be used in other regions. On the national level, this crucial data foundation can be obtained globally from FAOSTAT's Commodity Balance Sheets (FAOSTAT, 2021, CBS).⁸ The CBS also represent a core backbone of FABIO and thus provide a useful reference point for the required supply and use data on the subnational level.

The CBS contain detailed and balanced data on the physical supply and use of primary and processed agricultural products within each country. For the soy supply chain, the relevant included commodities are raw soybeans and their two main derivatives, soybean oil and cake. Table 1 shows the balances of these commodities for Brazil in the year 2013. The supply side is given by domestic *production* and *imports*, while the use side consists of the categories *food*, *feed*, *processing*, *seed*, and *other (non-food)* uses.⁹ Moreover, the table contains the category *stock addition* on the use side, where a positive value indicates a net increase in stocks in the given year, while a negative value indicates a decrease (i.e., a stock withdrawal).¹⁰

Table 1: Commodity Balance Sheet for the Brazilian soy complex in 2013 (kilotons)

	Supply		Use						
	Production	Imports	Exports	Processing	Feed	Food	Other	Seed	Stock Addition
Soybean	81699.8	283.0	42796.5	35513.2	637.0	724.6	0.0	1211.5	1100.0
Soy Oil	7077.0	5.8	1380.6	0.0	0.0	2712.7	2989.6	0.0	0.0
Soy Cake	27622.0	10.6	13333.6	0.0	14299.1	0.0	0.0	0.0	0.0

Source: FAOSTAT (2021)

The CBS of these three products captures the grand majority of the Brazilian soy complex. The compilation of a database covering all relevant supply and use categories on the *subnational* level is therefore the first key step of the modeling process. Once this database

⁸Note that in a recent revision of the FAOSTAT database, the Commodity Balances have been split into Food Balances (for food commodities) and Commodity Balances (for non-food commodities). The soy-related products soybeans and soybean oil are now part of the Food Balance, while soybean cake is contained in the Commodity Balance.

⁹Note that not all use categories are relevant for each soy commodity: Only soybeans are used as a processing input (being transformed into oil and cake) and as a seed for following growing cycles. Human food consumption comprises raw soybeans and soybean oil, while only soybeans and soybean cake are fed to livestock. Other uses can refer to various non-food purposes. In the case of soy, this category is only relevant for soybean oil, which can be used as an industrial raw material and for the production of biofuels (Johnson, 2008, Schmitz et al., 2008).

¹⁰This variable was derived from the *stock variation* variable contained in the CBS, in which a positive value indicates net decreases in the stock of the given commodity in the given year (i.e., a withdrawal of stocks accumulated in previous years, effectively increasing the current supply), while a negative value corresponds to a net increase, i.e., not consuming the current supply but using it to increase the stocks that can be consumed in subsequent years (FAOSTAT, 2022a). The FAO originally defines stock variation as a supply item, while in FABIO, this item is inverted and added to the use side as *stock addition*, which is standard practice in input-output accounting (Miller and Blair, 2009). I follow this convention, consequently interpreting positive stock addition values as increases in stocks and negative values as decreases in stocks.

is complete, supply can be allocated to use across subnational regions in a spatially explicit manner. The values given in the CBS for Brazil provide the national reference values to which the subnational data needs to be aligned.

However, not all of the required data is readily (and freely) available from public sources on the municipal level, which is the subnational level of analysis in this study. Table 2 gives an overview of the source data availability at this level of detail. Green cells indicate that the data is directly available from the public sources specified within the respective cell. This is the case for the production of soybeans, which is available from the Brazilian Institute of Geography and Statistics' (IBGE) Municipal Agricultural Production data (IBGE, 2022a), and for the trade of all three soy commodities, provided by the Brazilian foreign trade database Comex Stat (Comex Stat, 2022a). Blue cells indicate that the data is not directly available and needs to be estimated or derived from other (subnational) proxy data. Grey cells imply that this category is not relevant for the given commodity, as the national totals given by the CBS for Brazil (table 1) are zero.

Table 2: CBS data overview on the municipal level

	Supply		Use						
	Production	Imports	Exports	Processing	Feed	Food	Other	Seed	Stock Addition
Soybean	IBGE	COMEX	COMEX						
Soy Oil		COMEX	COMEX						
Soy Cake		COMEX	COMEX						

available
 estimate
 not relevant

Below, I describe the process of compilation, estimation and harmonization of all these data categories in physical quantities (tonnes). For the data that needs to be estimated, I use the best publicly available proxy information at the highest spatial resolution possible. Table 3 outlines the data and methods used for the estimation of each item. As all data is required on the level of Brazil's municipalities, appendix A.1.1 gives an overview of the structure and specifics of these administrative units and provides some descriptive statistics on them, including all CBS items as compiled or estimated below.

Table 3: Data and estimation procedure for missing CBS items on the municipal level

Item	Estimation procedure/proxy data	Data source
Processing use (bean)	State-level processing capacity allocated to individual processing facilities	ABIOVE (2013)
Production (oil & cake)	Translated from processing use via average national conversion rate derived from CBS	FAOSTAT (2021)
Food use (bean & oil)	State-level consumer baskets and municipal population	IBGE (2018, 2022c)
Feed use (bean & cake)	Municipal livestock numbers by species, further disaggregated into management systems using gridded distribution maps, and system-specific feed ratios	FAO (2018), Gilbert et al. (2018a,b,c), IBGE (2006, 2022b), Robinson et al. (2018)
Other use (oil)	Production capacity and input mix of individual biodiesel plants	ANP (2013)
Seed use (bean)	Fixed share of municipal soybean production, according to national seed use share derived from CBS	FAOSTAT (2021)
Stock addition (bean)	Storage capacity of individual grain storage facilities	IBGE (2014)

3.1.2 Soybean production

The production of raw soybeans is one of the necessary supply items that are directly available at the municipal level in Brazil. As part of the IBGE's comprehensive agricultural statistics, the Municipal Agricultural Production database includes survey-based data on production volume and value as well as planted and harvested area for a wide range of temporary and permanent crops (IBGE, 2022a).¹¹ I use the soybean production volume in tonnes as the first key supply item, representing the origin of the soy supply chain.

For industrial crop derivatives like soybean oil and cake, production data is not readily available at the municipal level. As the production of these goods is a direct result of the processing use of soybeans, the estimation of these items is jointly described further below in section 3.1.4.

As visible in table A1 in appendix A.1.1, there are 1963 municipalities that reported soybean production in 2013. Summing over all municipalities yields a total national production very close to the value reported in the CBS (table 1).¹² In a final harmonization step described in section 3.1.10, all supply and use items at the municipal level are re-scaled proportionally to match the national CBS value on aggregate. Figure C1 in appendix C.1 maps the soybean production volumes by municipality, showing that production is concentrated in the central and southern regions of the country.

3.1.3 Imports and exports

The availability of municipal foreign trade data from [Comex Stat \(2022a\)](#), the foreign trade database of the Brazilian Ministry of the Economy, represents a major facilitating factor in the subnational modeling of commodity flows from the point of production to the point of export handling. The municipal foreign trade data covers monthly imports and exports based on official declarations by exporters and importers.¹³ The data is classified by the traded goods, the country of origin/destination and the municipality where the importing/exporting company has its tax domicile.

By using this data for the transport modeling framework, I implicitly assume that goods are transported to the municipality where the respective exporting company has its tax domicile (before being hauled to their export destination). This implies that the tax residences of exporters coincide with logistic hubs physically handling the cargo. I acknowledge that this assumption might not always be realistic, especially when the tax locations of such companies are located distant from production regions (e.g., in commercial centers).¹⁴ However, plotting the soybean export volumes by municipality (see figure C1 in the appendix) reveals

¹¹Besides soybean, these also include other heavily traded crops such as oil palm, coffee, sugar cane or cottonseed.

¹²Note that the summary statistics in table A1 are already harmonized with the national CBS.

¹³In contrast to commercially available customs declarations, however, the Comex data does not retain information on these traders.

¹⁴A similar issue is also mentioned in the documentation of Trase ([Trase, 2018a, 2020](#)), which uses taxation records to link per-shipment trade data to individual companies and their assets. When the exporting companies are located far away from production regions, the soy may not actually pass these locations and flows can thus partly not be traced with reasonable certainty, leading to a share of flows with "unknown origin". Since the Comex data is also based on tax records, it suffers from a similar issue. Unlike Trase, however, I do not use fee-based closed data to address these potentially problematic export flows, thus creating a likely source of inaccuracy in the results.

that most exporting companies are indeed located close to the soybean cultivation regions or, if not, along exporting routes (i.e., in port cities or regional transport hubs). The same holds for soybean oil and cake, for which clear proximity (or overlap) of exporting and producing municipalities is visible (figure C2 and C3). I cautiously regard this as a justification for the usage of these locations as a proxy for locations that are actually passed by the exported goods along their export routes. This means that the subnational transport optimization model in section 3.2 does not need to model export flows to the points where they ultimately leave the country (e.g., ports, custom stations), but only to the municipality that hosts the exporting company. From this point onward, the respective export destinations and volumes are known. This also avoids potential bottlenecks within the model caused by major ports that channel significant parts of the total export volume.¹⁵

The three relevant soy products are contained as distinct goods in the Comex database. The data is pre-processed and harmonized with the country sample of FABIO as described in appendix A.1.2. National aggregates by commodity again match well with FAOSTAT CBS values for 2013 (see also section 3.1.10).

It is important to note that the import and export data at this stage only represent the foreign trade with countries and regions *outside* of Brazil. However, from the perspective of municipalities, the trade with other municipalities also represents imports and exports. The determination of these *subnational* trade flows is the task of the spatial optimization framework described in section 3.2.

3.1.4 Soybean processing use and production of soybean oil and cake

The industrial processing of soybeans is done in the crushing industry, which extracts oil from the raw beans and produces soybean cake (also called crunch or meal) as a high-protein by-product (Goldsmith, 2008b).¹⁶ Other common soybean derivatives such as soy sauce, soy milk or tofu are produced and consumed only in small quantities in Brazil (Landgraf, 2010) and are thus not considered here.

To estimate the soybean processing use and the corresponding oil and cake output at the municipal level, I make use of data on state-level processing capacities and locations of individual processing facilities published by the Brazilian Association of Vegetable Oil Industries (ABIOVE, 2013). State-level capacities are equally allocated to facilities within each state¹⁷, and aggregate capacities per municipality are then used to split the national processing use (as defined by the CBS) across municipalities. Details on this procedure are described in appendix A.1.3.

As shown in table A1, soybean processing is relatively concentrated among 69 municipalities, with an average annual processing use of 102.6 kt of soybeans. Figure C3 shows that processing industries are mainly located close to the soybean production centers in the central and southern states.

¹⁵Flows that pass such bottlenecks will be allocated proportionally to all destinations of the respective port, introducing uncertainty.

¹⁶Goldsmith (2008b) notes that a large portion of the value derived in the processing step is actually embodied within the cake, making soybeans a protein crop rather than an oil crop.

¹⁷This is a somewhat coarse but necessary assumption in the absence of facility-level capacity data. Precision could only be improved by investigating individual firms.

The corresponding municipal output of soybean oil and cake is then obtained by deriving average (mass-based) conversion factors from the CBS (i.e., the average amount of oil and cake derived from one unit of processed soybeans) and applying them to the municipal processing use. This also assumes that a fixed mass share of the processed soybeans is lost as waste. Based on this, a so-called mass-based commodity-equivalence factor¹⁸ can be derived, which indicates how much soybean biomass is – on average – embodied in one unit of oil and cake. This factor will be used in section 3.3 to convert flows of oil or cake back to equivalent amounts of raw soybeans (for details see appendix A.1.3).

3.1.5 Feed use

The use of feed crops in the livestock sectors represents another type of intermediate industrial use, as the contained energy is transformed into livestock biomass. Soybean cake and – to a lesser extent — also soybeans are important components of livestock diets in Brazil, especially for pigs and poultry (Caro et al., 2018, Kuepper et al., 2019, Smaling et al., 2008, Zani, 2014). According to Brazil’s CBS, more than half of the soybean cake produced was used domestically as livestock feed in 2013, with the remainder being exported. For soybeans, feed use amounted to less than 1% of production.

As open data on crop-specific feed intake in livestock sectors is not available on a sub-national scale for Brazil, this use item needs to be approximated using municipal animal numbers and typical feed rations for each relevant species. By doing so, one can take into account the differences in the livestock composition across municipalities. However, it has been highlighted that feed composition does not only vary significantly between livestock species, but also between different production systems for the same species, which can strongly depend on regional socio-economic and ecological conditions (Robinson et al., 2011, Seré and Steinfeld, 1996). For example, in less densely populated regions with abundant grasslands, typical pasture husbandry of cattle will be preferred over more intensive production systems (Robinson et al., 2011). For a country as large and geographically diverse as Brazil, such differences should not be disregarded when estimating local feed use. To take this into account, I make use of three key datasets:

1. Municipal livestock numbers by species from the IBGE’s Municipal Livestock Production Survey (IBGE, 2022b) as base data
2. Raster data on the distribution of livestock species by production system from the FAO Gridded Livestock of the World (GLW) database (Gilbert et al., 2018a,b,c, Robinson et al., 2018), employed to disaggregate municipal species numbers into distinct systems
3. System-specific feed rations from the FAO Global Livestock Emission Assessment Model GLEAM (FAO, 2018) to estimate the soy feed intake of the respective livestock groups

The estimation procedure contains several steps, integrating a wide range of data on different scales and formats, and is described in appendix A.1.4. Based on the feed ration data from GLEAM and relevant literature on the Brazilian livestock sector, I include pigs and poultry, but also cattle and buffaloes as relevant species. The resulting feed use estimates

¹⁸Such a factor is also used in Trase (2020).

reflect the entire livestock population as well as the location-specific agricultural conditions of each municipality. Figures C2 and C3 in the appendix show that the feed use of soybean cake and beans is spread throughout the country, but also highlight the hotspot regions of livestock farming in the central Cerrado region and the south of the country.

The extensive reliance on FAO data with global coverage (i.e., GLW and GLEAM) has the key merit that this method could essentially be applied to any country or region worldwide, even in the absence of subnational livestock numbers.

3.1.6 Food use

In Brazil, soybean oil is the main soy product for human food consumption. Roughly half of the domestically consumed soybean oil is used for food purposes (see table 1), satisfying 90% of the domestic demand for consumer vegetable oils (Belik, 2020). Aside from oil, a relatively small volume of raw soybeans is also used for food purposes, accounting for less than 2% of domestic soybean use.¹⁹

To take account of regional differences in nutritional behavior, I make use of the IBGE's Consumer Expenditure Survey (IBGE, 2018), which contains data on annual per-capita purchases of soybean oil differentiated by state.²⁰ Assuming that soybean food consumption follows a similar regional pattern as the consumption of soybean oil, the municipal food use of both products is estimated by allocating the national CBS values to individual municipalities according to municipal population numbers, weighted by the state-specific per-capita soybean oil purchases. A more detailed description of this can be found in appendix A.1.5

Intuitively, figures C1 and C2 show that food use is strongest in the population centers of the country (i.e., state capitals, major cities along the coast).

3.1.7 Other use

The non-food industrial use of soybean oil in Brazil accounts for the second half of domestic use of this commodity. Although soybean oil has a wide range of industrial applications (Schmitz et al., 2008), the only significant industrial demand in Brazil comes from the biodiesel industry (Belik, 2020, Kuepper et al., 2019). The domestic biodiesel industry has been fostered by the government over the last decades to promote the agricultural sector and reduce the dependence on diesel imports (Lima et al., 2011). Soybean oil is the main feedstock of the industry, accounting for 70% of feedstock inputs (ANP, 2013).²¹ Detailed data on the production capacities of individual biodiesel facilities and average macro-regional feedstock input mixes is published by the National Agency of Petroleum, Natural Gas and Biofuels (ANP, 2013). As elaborated in appendix A.1.6, municipal soybean oil use in the biodiesel sector is then estimated by allocating the national CBS value to each municipality by its share in the aggregate feedstock-weighted production capacity.

¹⁹Note that within the FAO's commodity balance statistics, the item "Soyabeans" comprises not only raw soybeans, but also soy sauce, soy paste and soy curd. However, the consumption of these and other soy-based food products (such as soy milk or tofu) are not considered separately in this study. These products have traditionally played a negligible role in Brazilian diets, although they have been gaining ground in recent years (Landgraf, 2010).

²⁰The data shows significant regional differences, ranging from 2.5 kg in Paraíba to 7.2 kg in Goiás (with a median of 4.2 kg).

²¹Other notable feedstocks are beef fat and cottonseed oil.

As can be expected, mapping the municipal use of soybean oil in the biodiesel industries (figure C2) shows that these industries also tend to locate in the proximity of soybean processing (i.e., oil-producing) industries.

3.1.8 Seed use

A share of soybean production is used to produce seed material for the following growing cycle. Taking this share from the CBS, the municipal seed use is assumed to be proportional to local soybean production levels (see also appendix A.1.7).

3.1.9 Stock addition

Stock additions refer to the net increase in stocks of a given commodity at the end of the year, compared to the previous year. According to the CBS (table 1), soybeans had a stock increase of 1.1 Mt (~1% of production) in 2013. The storage can either be done on farms or in local storage hubs, collecting harvests from multiple producers for further transportation. According to [Maia et al. \(2013\)](#), 87% of soybean storage is located outside of farms.

To spatially allocate the net increase in stored soybean volumes, geospatial data on the location and capacity of grain storage units from the IBGE's transport logistics database ([IBGE, 2014](#)) is used. Aggregate storage capacities per municipality are taken as a proxy for soy storage demand (see appendix A.1.8 for details).

3.1.10 Final harmonization and balancing

In a final step, all municipal data is harmonized to match the national aggregates for all CBS items and balanced to ensure equality of supply and use on the aggregate level.²² For the soybean production data and the trade data of all three soy items, small discrepancies with the national CBS are corrected by re-scaling all values proportionally (see appendix A.1.9 for more details). Minor pre-existing imbalances between total supply and use in the national CBS are adjusted via the stock addition item.

3.2 Modeling subnational flows

With the subnational database completed, the next step is to allocate subnational supply to use in a spatially explicit manner. More specifically, this entails modeling the *flows* of soybeans as they move from local production (or imports) along the subnational supply chain to local use, either being consumed, transformed or exported abroad. The same holds for the two transformation products oil and cake.

To this end, a transport optimization framework is employed, conducting this allocation under the assumption of cost-optimal behavior of the involved agents. Such optimization problems are commonly used in material flow modeling (e.g. [Godar et al., 2015](#), [Péra et al., 2019](#), [Smith et al., 2017](#)), but can vary significantly regarding their level of complexity and

²²Note that on the *municipal* level, supply and use are not yet balanced at this stage (i.e., there can be excess use or excess supply) as subnational trade flows are not yet accounted for. This balance is established via the transport optimization in the next section.

incorporation of real-world features. To assess the influence of the transport setting on the model's outcomes, two "extreme" settings are tested here: First, as the simplest approach possible, a minimalist model based on Euclidean distance as a proxy for cost is set up. Second, as a more complex counterpart, a network-based model is constructed, incorporating multiple transport modes and real-world features of the Brazilian logistic network. The outcomes of both will be assessed in section 4.

3.2.1 A simple Euclidean model

As the most basic approximation of reality, one can allocate supply to demand solely based on the Euclidean distance between supplying and demanding municipalities. This simply minimizes the overall mass-weighted distance (tonne-kilometers) to be covered in order to satisfy the demand of all municipalities with the existing supply.

I assume that domestic use is always first satisfied with domestic supply. Hence it is only *excess supply* that is allocated via the model to satisfy the *excess use* of other municipalities.²³

The model is straightforwardly defined by the following components:

Sets

- $p \in \{bean, oil, cake\}$: the three investigated soy products
- $a, b \in \{1, \dots, M\}$: the full set of municipalities, with a standing for municipalities as suppliers and b as users²⁴

Parameters

- e_{ab} : Euclidean distance in kilometers between the capitals²⁵ of each supplying municipality a and using municipality b
- $\bar{u}_b^p$: excess use of each municipality by product, derived from total municipal use u_b^p and total supply s_b^p as

$$\bar{u}_b^p = \begin{cases} u_b^p - s_b^p, & \text{if } u_b^p > s_b^p \\ 0, & \text{otherwise} \end{cases} \quad (1)$$

- $\bar{s}_a^p$: excess supply of each municipality by product, derived as

$$\bar{s}_a^p = \begin{cases} s_a^p - u_a^p, & \text{if } s_a^p > u_a^p \\ 0, & \text{otherwise} \end{cases} \quad (2)$$

Variables

- x_{ab}^p : transport volumes in tonnes by product between each pair of supplying and using municipalities

²³This assumption is introduced for further simplification of the model but could easily be relaxed.

²⁴ b is an *alias* of a , containing the same municipalities. If there are two indices defined, they represent aliases in all sets defined here.

²⁵I assume that capitals best approximate the economic centers of each municipality. Locations of municipality capitals are taken from the [IBGE \(2010\)](#).

- T : total Euclidean tonne-kilometers traveled by all flows of the model across all three products, representing the overall "cost" of transport in this simple setting

Equations

The objective function now simply minimizes the overall tonne-kilometers for the transport of all products:

$$\min T = \sum_p \sum_a \sum_b e_{ab} x_{ab}^p \quad (3)$$

subject to

$$\sum_a x_{ab}^p \geq \bar{u}_b^p \quad (4)$$

$$\sum_b x_{ab}^p \leq \bar{s}_a^p \quad (5)$$

$$x_{ab}^p \geq 0 \quad \forall \quad a, b \quad (6)$$

Equation (4) ensures that the excess use of each municipality is satisfied, while equation (5) ensures that no more than the excess supply is offered. Equation (6) establishes non-negativity for all flows.

The solution of the model yields a matrix of inter-municipal flows $\mathbf{X}_{ab}^p$ of dimensions $(a \times b)$ for each product p .²⁶ Note that the diagonal of this matrix (i.e., self-supply flows from a municipality to itself) is empty at this stage, as the model only allocates excess supply. The self-supply will be added back to the subnational trade matrix in a further step in section 3.3.

3.2.2 A multi-modal network model

Due to the large distances between producing regions and consuming or exporting regions and the nature of soybeans as a bulk commodity, transportation is a major cost component in the Brazilian soybean system. Although these conditions would in general favor high-volume transport modes such as rail or barge, road transport by truck is still the dominant mode in Brazil (Fliehr, 2013, Oliveira et al., 2016). Railway and inland waterway infrastructures are comparatively underdeveloped, but play a significant and increasing role in specific export corridors (Oliveira et al., 2016, Pera et al., 2021, Péra et al., 2019).²⁷ The road network is relatively extensive, but road quality varies strongly, significantly affecting transport costs (Fliehr, 2013).

²⁶The model is solved in R, using the package *transport* (Schuhmacher et al., 2020). Like the multi-modal model below, it could however also be easily solved in GAMS.

²⁷According to Pera et al. (2021), the modal split between road, rail and barge for soy transport (in percent) was 75/20/5 in 2010, compared to 67/24/9 in 2019.

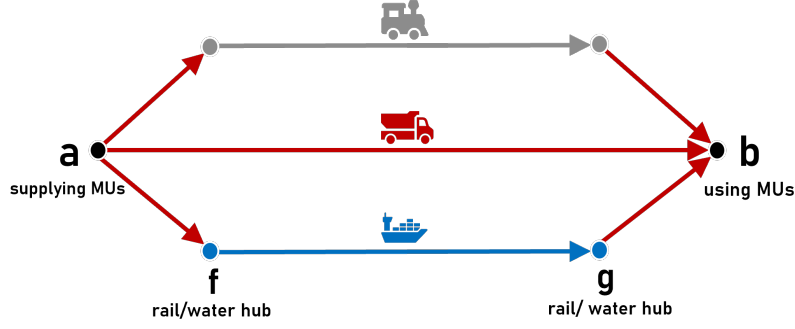


Figure 2: Schematic representation of transport options in the multi-modal model

Based on these characteristics of the Brazilian soy transport system, I construct a multi-modal transport optimization framework, incorporating the networks of all available transport modes and their relative cost. It assumes that agents involved in the supply chain will generally favor the cheapest transport option possible and thus allocates flows by minimizing the total transport cost, given the available supply, demand and transport routes. Moreover, it exploits highly specific cargo flow data on subnational rail and water transport from government sources (ANTAQ, 2021, ANTT, 2021). The model shares some basic features with the Euclidean model but adds flexibility by allowing agents to choose between pure road transport or inter-modal options that combine road with rail or waterway transport. It builds on existing work on soybean transport optimization modeling in Brazil (Cruz et al., 2010, Góes and Lopes, 2018, de Oliveira et al., 2021, Péra et al., 2019, Pinheiro and Caixeta Filho, 2010, Souza et al., 2020), which largely incorporates multiple transport modes, but usually focuses on identifying optimal export paths to ports. This work instead models *all* flows of soy within the country, including those required to fulfill domestic demand.

Figure 2 depicts the potential options available for hauling a product from a supplying municipality to a using municipality. It can either be achieved directly by truck via the road network or indirectly by switching to train or barge at any rail station or river port (rail/water hub) along the way. The first and last miles of such inter-modal routes are always carried out by truck, even if the respective supplying or using municipality hosts a rail station or river port. This is because the model (and the calculation of transport costs described further below) assumes that the starting- and endpoint of each flow is in the *capital* of the supplying and using municipalities, while the rail and waterway hubs are located at their proper location.²⁸ Importantly, the possible rail and waterway routes are constrained by existing cargo data as described below.

The optimization problem is defined as a linear model in GAMS. Its structure is defined as follows:

Sets

- products p and supplying/using municipalities a, b as defined above

²⁸The location of municipality capitals is defined by the coordinates given by the IBGE (2010). Rail stations and inland ports can be located very close to this coordinate, but will unlikely coincide with it. In such cases, the related road costs for the first/last mile will be very low.

- $f \in \{1, \dots, F\}$: *origin* rail and water hubs, which recorded domestic soy flows as a point of departure within the rail/waterway network according to the cargo data of [ANTT \(2021\)](#) and [ANTAQ \(2021\)](#)²⁹
- $g \in \{1, \dots, G\}$: *destination* rail and water hubs, which recorded incoming domestic soy flows

Parameters

- c_{ab} : road transport cost per tonne between supplying municipality a and using municipality b . Costs are assumed to not differ between the three soy products.
- c_{af} : road transport cost per tonne between supplying municipality a and origin rail/water hub f . They also include the transshipment costs of transferring the freight from trucks to trains or ships (switching costs).
- c_{fg} : rail/water transport cost per tonne between origin rail/water hub f and destination rail/water hub g ³⁰
- c_{gb} : road transport cost between destination rail/water hub g and using municipality b , including inter-modal switching cost. This differs from c_{af} as the set of destination hubs g (partly) differs from origin hubs f .
- $\bar{u}_b^p$: excess use per product, as defined above in equation (1)
- $\bar{u}_a^p$: excess supply of each municipality, as defined in equation (2)
- v_{fg}^p : recorded transport volume between origin rail/water hub f and destination rail/water hub g by product, as given by [ANTT \(2021\)](#) and [ANTAQ \(2021\)](#)
- e_b^p : export use (to other countries) of using municipality b by product
- ρ_b^p : processing use of using municipality b by product

Variables

- $\tilde{x}_{ab}^p$: direct road transport volumes by product between each pair of supplying/using municipalities (the tilde is used to indicate that this is only the direct part of the flows between a and b)
- x_{af}^p : road transport volumes by product between each supplying municipality and each origin rail/water hub
- x_{fg}^p : rail/water transport volumes by product between each pair of origin/destination rail/water hubs
- x_{gb}^p : road transport volumes by product between each destination rail/water hub and each using municipality
- Z : total absolute transport cost across all three products and modes

²⁹In the GAMS model, I define separate sets for rail and water hubs and also enter them separately into the model equations. For ease of notation, I combine them in one set here.

³⁰As it is impossible to transport from a rail hub to a water hub (and vice versa), the corresponding cost can be understood as infinite. In the GAMS model, such impossible links are already excluded by defining separate sets for rail and water hubs. See also footnote 29.

Equations

Objective function

$$\min Z = \sum_p \left(\underbrace{\sum_a \sum_b c_{ab} \tilde{x}_{ab}^p}_{\text{direct road route}} + \underbrace{\sum_a \sum_f c_{af} x_{af}^p + \sum_f \sum_g c_{fg} x_{fg}^p + \sum_g \sum_b c_{gb} x_{gb}^p}_{\text{inter-modal route}} \right) \quad (7)$$

The objective function now minimizes the overall transport cost for all products, which can be either transported via the direct road route or via the inter-modal road-rail or road-water routes, as illustrated in figure 2.

Constraints

$$\sum_a \tilde{x}_{ab}^p + \sum_g x_{gb}^p \geq \bar{u}_b^p \quad (8)$$

$$\sum_b \tilde{x}_{ab}^p + \sum_f x_{af}^p \leq \bar{s}_a^p \quad (9)$$

$$\sum_a x_{af}^p = \sum_g x_{fg}^p \quad (10)$$

$$\sum_f x_{fg}^p = \sum_b x_{gb}^p \quad (11)$$

$$x_{fg}^p \leq v_{fg}^p \quad (12)$$

$$\sum_g x_{gb}^p \leq \epsilon_b^p + \rho_b^p \quad (13)$$

$$\tilde{x}_{ab}^p, x_{af}^p, x_{fg}^p, x_{gb}^p \geq 0 \quad \forall \quad a, b, f, g \quad (14)$$

Analogous to the Euclidean model, equation (8) ensures that the excess use of each municipality is satisfied, while equation (9) ensures that no more than the excess supply is offered. For the inter-modal routes, equation (10) guarantees that everything that is transported from supplying municipalities to origin rail/water hubs is also hauled further to destination hubs. In a similar way, equation (11) makes sure everything is passed on from destination hubs to using municipalities. Importantly, equation (12) limits the possible rail and water transport flows to the volumes observed in the official cargo data, ensuring that the alternative modes are not used in excess compared to what is observed in reality. It imposes an upper limit (and not equality) because the municipal export data from Comex refers to the domiciles of exporting companies, which are often not located in the municipality from which the freight actually leaves the country (in most cases those with seaports) but in other (inland) locations along the export route. In such cases, the export use is allocated

to municipalities that do not correspond to the actual end-point of the subnational export route. As the "invisible" transport from inland exporting municipalities to eventual export hubs could be conducted by train or inland waterway (as depicted by the cargo data), the constraint is formulated as an upper limit. Related to this, equation (13) limits the inter-modal transport option to the export and processing use of each using municipality. This is based on the observation that train and barge are only used for export-oriented soy transport, while domestic demand is (almost) exclusively covered by road transport (Pera et al., 2021). Finally, all transport flows are again constrained to be non-negative (equation (14)). Appendix A.2.1 gives some details of the implementation of the model in GAMS.

Unlike the Euclidean model, which returns a single matrix of flows between a and b , the different flows of the multi-modal model across different modes need to be reconciled. Aggregate flows between each pair of municipalities across all possible routes are computed as

$$x_{ab}^p = \tilde{x}_{ab}^p + \sum_g \left(\sum_f r_{af}^p r_{fg}^p \right) x_{gb}^p, \quad (15)$$

where $r_{af}^p = \frac{x_{af}^p}{\sum_a x_{af}^p}$ represents the relative share of supplying municipality a in all flows arriving in origin rail/water hub f and r_{fg}^p analogously represents the relative share of origin rail/water hub f in all flows arriving in destination hub g .³¹ More conveniently, equation (15) can also be expressed in matrix notation, defining $\mathbf{X}_{ij}^p$ as the matrix with elements x_{ij}^p of flows between node i and j , and $\mathbf{R}_{ij}^p$ as the same matrix with columns re-scaled by their sums (i.e., each element r_{ij}^p representing the relative share of origin-node i in all flows arriving in destination-node j):

$$\mathbf{X}_{ab}^p = \tilde{\mathbf{X}}_{ab}^p + \mathbf{R}_{af}^p \mathbf{R}_{fg}^p \mathbf{X}_{gb}^p. \quad (16)$$

The matrix $\mathbf{X}_{ab}^p$ with elements x_{ab}^p now contains all flows of product p between all possible pairs of municipalities (equivalently to the Euclidean distance case).

Computing mode-specific transport costs

Mode-specific transport costs c represent a key ingredient in the described model. As can be seen in figure 2 and the formal definition above, the model requires road transport cost matrices between each pair of municipalities (c_{ab}) and between each municipality and rail/water hub (c_{af}, c_{gb}). Moreover, railway and waterway shipping cost matrices between each origin and destination rail/water hub (c_{fg}) are needed. For all three modes, I use detailed public network datasets to calculate shortest-path distances between all relevant points. The road network is taken from OpenStreetMap (Geofabrik, 2018) due to its high level of detail and global coverage, while rail and waterway networks are taken from official government

³¹Note that this calculation implies that all outflows of each origin and destination hub consist of a proportional mix of all its incoming flows across their origin. This represents a model bottleneck and might not be fully realistic due to temporal differences in individual flows and individual trading contracts. However, in the absence of trader-specific data, this assumption is inevitable.

sources (ANTT, 2020, DNIT, 2021).³² Shortest-path calculations are based on rasterized versions of these networks. This method has been applied in similar contexts by Victoria et al. (2021) and Vera-Diaz et al. (2009) and has the advantage that start-/endpoints do not necessarily need to lie on the network itself. This is particularly useful for calculating road distances to/from municipalities that are not connected to any road in the data. To account for differences in road quality, speed limits of roads are taken as a proxy for road quality, effectively weighting the travel distances. Transport costs (per tonne) are then obtained by multiplying (weighted) shortest-path distances by mode-specific average costs (per tonne-kilometer). A detailed description of the involved working steps and data sources for each transport mode can be found in appendix A.2.2.

Assessing model sensitivity

The mode-specific cost values used to transform distances into costs represent exogenous input parameters and are taken from the literature. However, such cost estimates vary significantly across sources, and the particular values chosen can have a considerable influence on the model's results. Thus, instead of using a single cost value per mode, I define ranges based on a literature review. The model's sensitivity to these key parameters is then assessed by running Monte-Carlo simulations, solving the model 1000 times using random picks from these ranges. Details on the surveyed literature, the simulation procedure and its results are given in appendix A.2.3. Results reveal that the large majority of transport flows exhibit only minor variation across simulations, indicating that the model is relatively insensitive to the cost parameters. This can likely be attributed to the high specificity of data-driven constraints, especially equation (12). For all further steps below, the average of each inter-municipal flow across all simulations is used as the core result of the multi-modal model.

3.3 Linking subnational to international flows

In the third key step of the framework, the inter-municipal flows of soy products (either from the Euclidean or the multi-modal model) are traced further to the countries of first import (i.e., the countries that directly import soybeans, soybean oil or soybean cake from Brazil).³³ Moreover, the flows of soybean oil and cake can be traced back to the origin of the embodied soybeans.

3.3.1 From subnational flows to importing countries

Using the municipal export data, it is straightforward to connect the subnational flows to the export destination countries. The goal here is to obtain the flows of each product

³²The data is selected to depict the state of the transport network at the time of analysis (in this case the year 2013) as accurately as possible. Historic OSM data from the end of 2014 is used to allow some time lag in the addition or modification of new roads in the community-based platform. The rail network data from ANTT represents the official publicly granted railway mesh from the year 2016, which is the earliest year available on the agency's website. The waterway network from DNIT is based on the national transport plan of 2014. For more details on the underlying data, see appendix A.2.2.

³³Note that international trade often involves re-exports, e.g., from the main European importers like the Netherlands to other European countries. These are not considered here but can be included in a further step, either by using bilateral trade data as described by Kastner et al. (2011) or by fully integrating the flows into an MRIO framework as outlined in section 5.2.1.

from the producing municipalities (soy cultivation municipalities for soybeans, processing municipalities for soybean oil and cake) to the importing countries.

To achieve this, it is first necessary to re-enter the self-supply of municipalities by filling the diagonal of the subnational trade matrices with self-supply values so that

$$\text{diag}(\mathbf{X}_{ab}^p) = \mathbf{s}^p - \bar{\mathbf{s}}^p, \quad (17)$$

where $\mathbf{s}^p$ and $\bar{\mathbf{s}}^p$ represent the vectors of total and excess supply respectively. The row sums of $\mathbf{X}_{ab}^p$ (i.e., total outflows) then correspond to the total supply of product p in each municipality, and the column sums (total inflows) analogously equal the total use per municipality.

Next, keeping in mind that the total supply per municipality s_a^p can consist of both domestic production and (in some cases) imports from other countries, the subnational trade matrix is adjusted to only contain domestically produced flows. Denoting domestic production of municipality a as π_a^p , let θ^p be the vector of municipal domestic supply shares with elements $\theta_a^p = \frac{\pi_a^p}{s_a^p}$. The adjusted flows of *domestically produced* products are then obtained as

$$\dot{\mathbf{X}}_{ab}^p = \hat{\theta}^p \mathbf{X}_{ab}^p, \quad (18)$$

where $\hat{\theta}^p$ represents the diagonalized form of θ^p .

Assuming that all inflows into a municipality are distributed proportionally to each use category (exports, food, feed, etc.), the subnational flows of each product can now be linked further to the destination countries K of municipal foreign exports. For this purpose, I define the municipal foreign export matrix $\mathbf{E}_{bk}^p$ with elements e_{bk}^p representing the exports of product p from municipality b to importing country k according to the Comex municipal export data. To additionally consider flows that are consumed domestically, I interpret the domestic consumption of product p (i.e., all use categories except exports) in each municipality as "exports to Brazil".³⁴ Therefore, the set of destination countries K contains all export destinations of Brazilian soy products, plus Brazil itself as an aggregate representation of domestic consumption. For raw soybeans, this domestic consumption excludes the processing use, as the processing industries are part of the modeled soybean supply chain and the flows of processed beans are embodied in the flows of oil and cake (as considered below in section 3.3.2).

To link the subnational with the international flows, I first derive a matrix of domestic source shares as

$$\dot{\mathbf{R}}_{ab}^p = \dot{\mathbf{X}}_{ab}^p (\hat{\mathbf{u}}^p)^{-1}, \quad (19)$$

³⁴In principle, one could treat every municipality as a distinct "importer" of domestic use, but for convenience they are aggregated here. This is also done in the Trase database.

where $\hat{\mathbf{u}}^p$ is the diagonalized vector of total municipal use of product p .³⁵ The matrix of flows from the supplying municipality to the country of first import is then simply given by³⁶

$$\mathbf{O}_{ak}^p = \hat{\mathbf{R}}_{ab}^p \mathbf{E}_{bk}^p. \quad (20)$$

Element o_{ak}^p of this origin-to-importer matrix represents the flow of product p from the producing municipality a to the importing country k .

3.3.2 Unifying the soybean supply chain

From the perspective of material flow analysis, it is of interest to unify the subnational soy supply chain by tracing the raw soybeans embodied in soybean oil and cake flows back to their origin. To achieve this and thereby also make the results comparable to the data of Trase, two additional steps are necessary. First, oil and cake flows (O_{ak}^{oil} and O_{ak}^{cake}) are linked back from the processing municipalities to the producing municipalities of embodied soybeans via the inter-municipal soybean source matrix $\hat{\mathbf{R}}_{ab}^{bean}$ as³⁷

$$\mathbf{O}_{ak,orig}^p = \hat{\mathbf{R}}_{ab}^{bean} \mathbf{O}_{bk}^p \quad \text{for } p \in \{\text{oil, cake}\}. \quad (21)$$

Second, these flows are expressed as raw soybean equivalents by multiplying them with the mass-based equivalence factor derived from the national CBS in section 3.1.4 (and described in detail in appendix A.1.3), equal to 1.023:

$$\mathbf{O}_{ak,orig}^{p,eq} = 1.023 \mathbf{O}_{ak,orig}^p \quad \text{for } p \in \{\text{oil, cake}\}. \quad (22)$$

The equivalence factor indicates that 1 kg of soybean oil and cake on average embodies 1.023 kg of soybeans due to processing losses of 23 g.³⁸ Using this factor ensures that the oil and cake flows are coherently transformed into embodied raw soybeans (i.e., in line with the estimation of municipal oil and cake production in section 3.1.4). This, in turn, allows the summation of the soybean flows and the soybean-equivalent flows of oil and cake per producing municipality and destination country to form an aggregate soybean flow matrix:

$$\mathbf{Q}_{ak}^{bean} = \mathbf{O}_{ak}^{bean} + \mathbf{O}_{ak,orig}^{oil,eq} + \mathbf{O}_{ak,orig}^{cake,eq}. \quad (23)$$

Due to the use of the correct equivalence factor and the inclusion of Brazil as the destination of non-export flows, the row sums of this matrix are equal to the total soybean production per municipality as initially given by the IBGE agricultural production data. The

³⁵The matrix $\hat{\mathbf{R}}_{ab}^p$ can be seen as a municipal "input coefficient matrix", giving the relative shares of row-municipalities as a source of product p used in column-municipalities. Note that the column sums of this matrix are lower than one if a municipality uses imports from abroad.

³⁶Note that this is similar to the approach described by Godar et al. (2015), but excludes the further tracing of flows via bilateral re-exports.

³⁷This again assumes that all uses of soybeans in a municipality (exports, processing, etc.) share the same spatial source structure.

³⁸This factor is slightly lower than the one used by Trase of 1.031 and therefore assumes fewer processing losses.

representation of flows according to $Q_{a,k}^{bean}$ now also corresponds exactly to the data structure of Trase, allowing a direct comparison of the datasets.

Note that the linkage steps described in this section would not be necessary when the subnational supply chain is integrated into the input-output framework of FABIO. In this case, the simple trade-linkage would be replaced by a full nesting into the multi-sectoral and multi-regional setting of the MRIO (see also discussion section 5.2.1).

4 Benchmarking

The need for subnational supply chain data for the refinement of footprint analyses has been set forth extensively (e.g. [Godar et al., 2015](#), [Moran et al., 2020](#), [Sun et al., 2018](#)), but as for international trade data, the accuracy of such fine-scale assessments strongly hinges on the accuracy of the modeled flows. Therefore, it is useful to contextualize the results of the proposed framework by benchmarking them against existing state-of-the-art data. As the main reference, I use the supply chain data of the Trase platform ([Trase, 2022](#)), which has been at the forefront of the methodological development in this field. Although a model itself, I regard this database as a key benchmark due to its extensive use of microdata and its sophisticated modeling approach. At the scope and scale relevant for spatially explicit consumption-based footprint analysis, and in the absence of systematic real-world data, it is currently simply the most accurate data available.

Additionally to this key reference, it is useful to specify a benchmark representing the "low end" of the complexity spectrum for the subnational allocation of trade flows. The obvious choice for this is the pure downscaling approach, which simply assigns national exports proportionally to subnational regions based on their production levels. This approach has been applied in several large-scale footprint analyses in the absence of subnational supply chain data ([Bruckner et al., 2019](#), [Hoang and Kanemoto, 2021](#), [Kanemoto et al., 2016](#)) and has the evident advantage of universal coverage without the need for modeling subnational flows. However, the usefulness of this approach regarding the assessment of environmental responsibilities has been contested ([Croft et al., 2018](#), [Parra Paitan and Verburg, 2019](#)).

The "complexity range" given by these two benchmarks allows a tentative investigation of two key questions: 1) How close do the results of my open-source framework come to the microdata-based Trase model, compared to a pure downscaling, and 2) how much added value does the more complex multi-modal model give to the simpler Euclidean approximation?

I emphasize that the benchmarking exercise does not intend to give a judgment about which model is "better" or "more true". Such a clear-cut verdict is impossible without systematic real-world validation data and is also impeded by some fundamental differences in the data described below. Rather, the aim of this exercise is to give some insights into the uncertainty related to fine-scale footprinting and the value of open-data-based approaches within this complex topic. Such a comparative evaluation of different approaches to fine-scale footprinting is so far lacking from the literature. Moreover, the benchmarking can also provide some useful information regarding the level of accuracy when using the proposed model framework in combination with IO data, as will be outlined in the discussion in section 5.

4.1 The benchmarks

4.1.1 Trase

The Trase project is a spatially explicit supply chain mapping initiative building on a modeling framework originally introduced by [Godar et al. \(2015\)](#). The platform has continuously expanded and currently contains subnational supply chain data for eleven commodities across ten countries.³⁹ Brazil's soy supply chain has been the front-row product with the most sophisticated methodology.

Trase's modeling approach is similar to the one employed here, integrating a wide range of subnational data on production, usage, logistics and trade. Its key differentiator is the usage of per-shipment trade data (customs records, bills of landing), which is linked to individual trading companies and their geo-localized export hubs (e.g., farms, storage, processing or wholesale facilities) via tax registration data ([Trase, 2018a](#)). From these logistic hubs, the flows are then traced to their likely origin (e.g., soy production municipalities) by means of a transport optimization model based on road network distance. The optimization minimizes transport distances while maximizing the sourcing from municipalities where the respective traders have relevant assets ([Trase, 2018a, 2020](#)).⁴⁰ Note that the model presented here is similar to Trase in that it models export flows via exporters' tax locations (via the Comex municipal export data). However, the diverse information on individual companies used in Trase crucially adds to the accuracy of results and reduces uncertainty. As a result, flows are not only traced to the producing municipality, but also to the trading companies and ports/custom stations they pass through.

Notably, there is a small share of flows associated with export hubs located far away from producing regions (e.g., wholesale facilities in São Paulo), which are assigned an unknown origin by Trase, as the determination of their source regions is highly uncertain.⁴¹ This complicates the direct comparison to Trase data to some extent, as the presented model assigns an origin to all flows, albeit with a higher degree of uncertainty. In all comparisons below, the Trase data excludes flows with unknown origin, since they have no counterpart in my model results. This issue would be worth further consideration, but this is beyond the scope of this thesis. Additionally, due to the usage of per-shipment trade data, the total exports by destination country do not correspond one-to-one to the official government data from Comex. Discrepancies vary across countries and are shown in table B1 in the appendix. These points prevent an unrestricted direct comparison of my model to Trase and need to be considered when interpreting the results.

Nevertheless, the Trase data can be regarded as a key benchmark, representing the most accurate available subnational supply chain data covering the whole country. It is

³⁹As of April 2022, the commodities covered are soy (Argentina, Brazil, Paraguay, Uruguay), palm oil (Brazil, Colombia, Indonesia), beef (Brazil, Paraguay, Colombia), shrimp (Ecuador, Indonesia, Peru), cocoa (Brazil, Colombia, Côte d'Ivoire, Ghana, Peru), coffee (Brazil, Colombia, Peru), corn (Argentina, Brazil, Paraguay), wood pulp (Argentina, Brazil, Colombia, Indonesia), chicken (Brazil), cotton (Brazil, Argentina), and pork (Brazil).

⁴⁰In this model, the trader-owned export hubs compete with domestic demand locations (processing facilities and oil refineries producing for the domestic market). Due to the usage of company data, the products destined for export and domestic demand can be separated already at the place of processing or storage.

⁴¹A similar issue likely also persists in the municipal export data of Comex, which also contains exports from large coastal cities (i.e., the exporter has its tax location in these cities), and it can not be guaranteed that the exports are actually handled in these locations. This issue is not addressed by the approach presented here, which increases potential uncertainties in the results.

not systematically validated against real-world data, but partially verified by company information (Trase, 2018a). All data used by Trase is publicly available, but company-related data (e.g., tax and custom records) are not free. As a consequence, the underlying data basis and calculations are not fully published, and therefore not directly replicable.⁴² The comparative exercise conducted here aims to shed some light on how (dis)similar the open-source model results are to those of Trase.

4.1.2 Pure downscaling

To create a pure downscaling benchmark, the national exports of Brazil (again including domestic consumption as "export to Brazil") are allocated proportionally to producing municipalities according to the share of their production in total national supply (production plus imports). For soybean oil and cake, the downscaling can be understood as a two-step process: First, exports are allocated proportionally to processing municipalities, and then from there – again proportionally – to soybean-producing municipalities.⁴³ When converting oil and cake flows to soybean equivalents, the total outflows per soybean municipality again correspond to their production volume.

4.2 Overall comparison

Before investigating results for individual destination countries, I assess the overall correlation between the different models, with Trase as the reference benchmark. The scatterplots in the first row of figure 3 plot origin-destination-specific flow volumes (in kilotonnes) of Trase (i.e., for each combination of municipality and importing country) against those of the multi-modal model, the Euclidean model and the pure downscaling. Flows that are zero both in Trase and the respective comparison model are not included. Additionally, the plots include a green line with the slope corresponding to Pearson's r correlation coefficient, as displayed on the bottom right together with its p -value⁴⁴. The dashed grey line has a slope of one. According to this direct global comparison, all three models show a similar intermediate association of about 0.7 with Trase. The multi-modal model only shows a negligibly better fit than the other two models, which both show the same correlation.

However, a direct (one-to-one) comparison might not be ideally suited if one is interested in comparing the spatial *patterns* of flows regarding their origin. For example, two models could show a similar pattern regarding the source regions of flows to a particular destination, but with slight displacements of exact origins (e.g., from neighboring municipalities). A direct comparison, which doesn't account for the spatial proximity of source municipalities, would identify such slight mismatches as major discrepancies, although a visual inspection might find that the spatial origins are very similar. For the comparison of spatial patterns, several methods have been developed, which typically depend on the definition of spatial

⁴²Trase does not publish the code files underlying their models. Data sources specific to each supply chain and logistic datasets for Brazil are published at <https://supplychains.trase.earth/>.

⁴³This is equivalent to downscaling oil and cake exports directly to the soybean-producing municipalities and adjusting for the national import share of these products.

⁴⁴The p -value is based on a Student t -test with $n-2$ degrees of freedom, assuming a normal distribution of both variables. Note that the square of Pearson's r is equivalent to the R^2 of a linear regression between the two respective datasets.

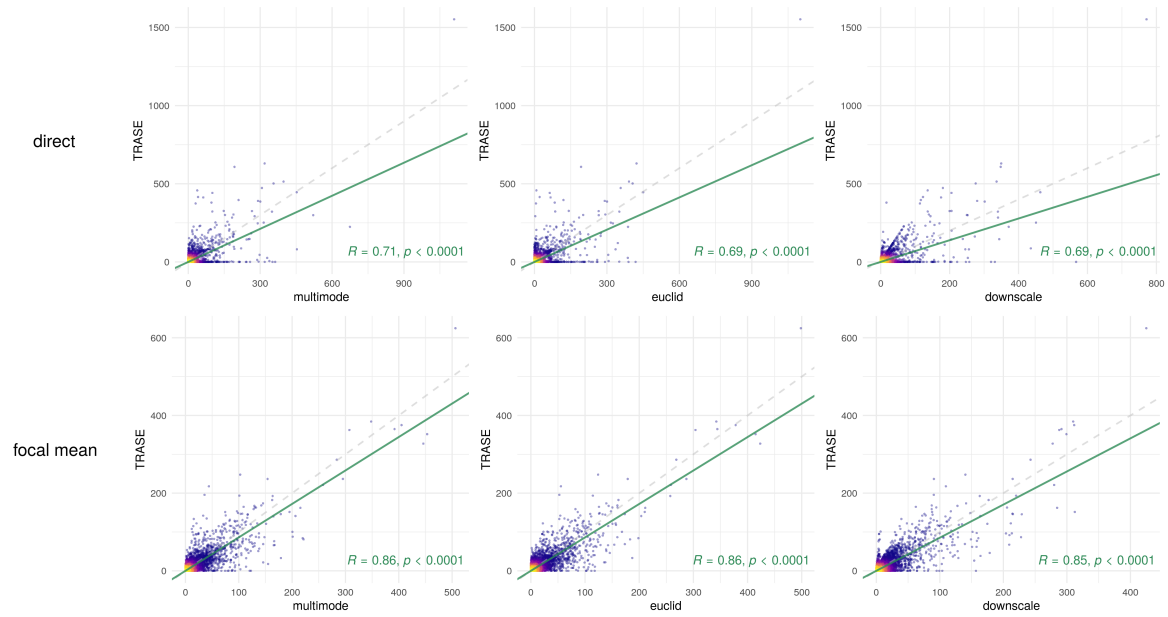


Figure 3: Correlation of different models with Trase. The top row plots individual flows directly against each other, while the bottom row plots focal means across neighboring municipalities. Each dot represents an origin-destination flow (or its focal mean), with colors giving kernel densities (bandwidths of 100 for the direct and 50 for the focal mean comparison). Green lines give the Pearson's r correlation coefficient, as indicated on the bottom right of each plot (including its p -value). Dashed lines have a slope of 1.

neighborhoods around the points of interest.⁴⁵ Here, I draw from these approaches rooted in spatial statistics and correlate the *focal means* of flows across the neighborhood of origin municipalities. I.e., instead of comparing flows one-to-one, I compute for each flow the mean of flows per destination across the neighborhood of the respective origin-municipality, including the origin-municipality itself.⁴⁶ As the spatial *neighborhood*, I define all municipalities bordering the respective municipality (also known as the "queen contiguity" case). As this choice is somewhat arbitrary, I also test an alternative definition of all municipalities within a radius of 100 km, with correlation results displayed in table B2 in the appendix. The

⁴⁵These methods include the multivariate Moran's I measure (also known as Anselin's LISA in the case of local spatial associations (Anselin, 2019, Anselin et al., 2002)), and a variant of it adapted to flow data (Tao and Thill, 2020), spatial cross-correlation Chen (2015), Lee's L statistic of bivariate spatial association (Lee, 2001), as well as moving window (Costanza, 1989) or fuzzy map comparison approaches (Hagen-Zanker, 2006, Negreiros et al., 2022). They differ in the statistical approaches taken, but share the feature that the cells or polygons of one map are typically compared to the spatial neighborhood of the counterpart cell/polygon in the reference map.

⁴⁶Note that this can include zero-flows if a neighboring municipality does not export to a given destination. Only municipalities that produce soy are included here. I.e., if an origin-municipality borders no other municipalities that produce soy, its neighborhood is empty. The approach taken here is similar to the bivariate Moran's statistic, with the difference that the spatial lag of *both* variables is used, and in-situ correlation is also considered. It is also similar to Lee's L , but without considering the univariate autocorrelation of each variable.

neighborhood is defined by a spatial weights matrix, where each bordering municipality (or each municipality within 100 km) and the target municipality itself get the same weight, resulting in a focal mean for each observation.⁴⁷

The correlation of these focal means is depicted in the bottom row of figure 3. The association of the benchmark models with Trase becomes stronger, indicating that the consideration of the spatial neighborhoods improves the fit. Strikingly, however, this is also the case for the pure downscaling, which again performs similar to the two models presented here. At first glance, and relating to the questions posed at the beginning of this section, these results may indicate 1) that the two subnational models do not give more accurate results than a simple downscaling and 2) that the more complex multi-modal model does not add value compared to the Euclidean-distance-based allocation. However, the overall comparison may mask important differences in the profiles of individual importing countries, which is generally of interest when tracing consumption-based footprints of individual nations. Thus, instead of looking at the opaque overall picture, country-level comparisons can give a more insightful perspective.

4.3 Comparison by destination country

To compare the models on the level of individual destination countries, the correlation analysis is repeated by destination, with results for the top 20 importers shown in table 4.⁴⁸

The results reveal marked differences across countries. For China, the main importer of Brazilian soy, all three models correlate strongly with Trase, with the downscaling again performing only slightly worse than the two subnational models. As China accounts for a large part of the total export flows⁴⁹, this high correlation also strongly affected the overall correlation results in figure 3 (and the bottom row of table 4). The high similarity of China's sourcing pattern in Trase to the pure downscaling case (both in the direct and focal mean comparison) suggests that China sources from *all* soy-producing regions in Brazil, with the relative spatial distribution similar to the distribution of production volumes. This is illustrated further in figure 4, which compares the sources of soy flows to China across the four benchmarks. Although the patterns in the multi-modal, Euclidean and downscaling results seem more dispersed compared to Trase (i.e., less clear-cut and spread out to a wider range of municipalities), the hotspot regions are markedly similar and correspond well with the production hotspots visible in the downscaling map. This is also visible in the scatterplots of focal means for China, displayed in the top row of figure 6.

The flows consumed domestically ("exports to Brazil") represent the second largest destination but show much worse accordance across the models.⁵⁰ This might be explained by the fact that the subnational domestic consumption (i.e., the domestic use items of the CBS) are estimated with different methods here compared to Trase. Based on Trase's documentation (Trase, 2018a, 2020), it is not possible to conclusively assess and compare the accuracy

⁴⁷Statistical comparisons of spatial data often suffer from spatially correlated residuals, which hamper statistical inference. An OLS model regressing Trase data directly on the subnational model results indeed shows spatially correlated residuals when conducting a Moran's I test. This autocorrelation is no longer present when the focal means are used instead.

⁴⁸Full results for all importing countries, including a separate set of results for the neighborhood definition of 100 km are displayed in table B2 in appendix B.

⁴⁹Note that the total exports to China (and most other countries) included in this comparison are lower in Trase compared to the Comex data used in this study, due to "unknown origin" flows in Trase being excluded.

⁵⁰A map comparison for Brazil is available in appendix C.2.

Table 4: Correlation between different model approaches with TRASE for the top 20 export destinations. The first two columns give export volumes according to TRASE and the multi-modal model in kt. Columns 3-5 give direct Pearson correlation coefficients between the respective model's flows and TRASE, and columns 6-8 give the same correlations for the focal means across neighbouring municipalities. Stars indicate significance of the correlation according to the p-value (≤ 0.001 : ***, ≤ 0.01 : **, ≤ 0.05 : *). For country abbreviations see table B1.

	exports (kt)		direct correlation			focal mean (queen) correlation		
	trase	comex	multimode	euclidean	downscale	multimode	euclidean	downscale
CHN	29253.1	32737.8	0.8201***	0.8203***	0.8156***	0.9045***	0.9071***	0.8941***
BRA	23922.6	24022.7	0.3691***	0.3539***	0.4139***	0.6506***	0.6724***	0.6991***
NLD	5344.8	5932.6	0.6806***	0.6087***	0.6515***	0.8227***	0.7779***	0.7719***
ESP	2173.2	2210.0	0.793***	0.8383***	0.3208***	0.8867***	0.9118***	0.3483***
THA	1647.7	2004.6	0.6375***	0.5995***	0.5773***	0.7696***	0.7777***	0.7322***
DEU	1602.5	1591.8	0.6284***	0.5467***	0.2758***	0.6446***	0.5941***	0.2656***
KOR	1499.3	1500.3	0.656***	0.5983***	0.2441***	0.7502***	0.7161***	0.2649***
FRA	1346.8	1723.0	0.533***	0.4369***	0.433***	0.6733***	0.6094***	0.5541***
VNM	950.8	1021.8	0.7162***	0.6776***	0.5438***	0.8431***	0.8184***	0.7224***
TWN	921.9	984.1	0.6119***	0.5879***	0.4503***	0.7347***	0.7374***	0.5229***
JPN	799.8	845.4	0.8831***	0.8756***	0.5849***	0.9394***	0.9282***	0.7478***
IRN	723.2	755.3	0.5592***	0.4111***	0.3781***	0.7269***	0.4978***	0.4198***
ITA	631.8	723.7	0.8288***	0.8328***	0.5944***	0.8774***	0.8866***	0.7166***
IDN	580.9	602.7	0.3253***	0.1794***	0.4305***	0.6249***	0.484***	0.5929***
GBR	540.9	638.0	0.6567***	0.5879***	0.542***	0.7825***	0.7555***	0.6568***
SVN	460.2	497.5	0.3893***	0.3374***	0.2948***	0.4813***	0.4419***	0.3668***
NOR	390.6	329.4	0.9684***	0.9649***	0.3416***	0.966***	0.9634***	0.3891***
SAU	307.0	398.4	0.7582***	0.6233***	0.2313***	0.85***	0.7861***	0.2885***
USA	294.6	324.0	0.8135***	0.7746***	0.4005***	0.9172***	0.8897***	0.4473***
MEX	275.9	286.2	0.8738***	0.876***	0.3671***	0.9596***	0.9392***	0.408***
Total	75681.52	81699.78	0.7073***	0.6919***	0.6944***	0.8628***	0.8637***	0.8525***

of its estimation of domestic demand compared to the approach used here.⁵¹ The fact that the domestic demand consists mainly of demand for processed soybean cake and oil (as opposed to exports, which often contain a large share of raw soybeans that are processed abroad) might be another complicating factor, as the relatively low number of processing facilities represents a bottleneck in the model and adds an additional layer of uncertainty.

For the Netherlands, Europe's largest importer of Brazilian soy and the second-largest importer overall, the moderate correlation of the multi-modal and Euclidean models, which is again not significantly higher than for the pure downscaling case, might be influenced by three factors: i) the sourcing of large amounts of soy from a wide range of regions (similar to China), ii) the large share of processed cake in soy imports (see table B1 in appendix B) and iii) the high share of "unknown origin" flows to the Netherlands in Trase (as also visible in table B1). However, looking at countries with lower import volumes in table 4, the following pattern is often observed: i) For most countries, the downscaling performs significantly

⁵¹The Trase documentation states that "the distribution of domestic demand is estimated by mapping the location and capacity (e.g., storage capacity, mill capacity, slaughtering capacity, etc., depending on the commodity of interest) of individual processing facilities in the country" (Trase, 2018a, p.8) and that "soy is domestically used almost exclusively for feed (crushing facilities) and for biofuels (oil refineries)" (Trase, 2020). This may suggest that Trase puts less emphasis on the estimation of domestic demand for soybean derivatives, such as the feed use of cake or the food and energetic use of oil. Such an estimation might not be necessary due to information on the export orientation of soy processors (i.e., whether a crushing/refining facility produces for the domestic market or the export markets).

worse than the two subnational models, ii) this is especially the case for countries with a higher share of raw soybean imports (which are not affected by the processing bottleneck) and iii) the multi-modal model generally performs slightly better than the Euclidean one. To visualize this, I take the example of Spain, the second biggest (direct) European importer of Brazilian soy.⁵² Figure 5 clearly shows that the multi-modal and Euclidean models perform much better in finding a similar source pattern to Trase than the pure downscaling. The corresponding scatterplots in the bottom row of figure 6, confirm these visual similarities. Similar outcomes can be observed for several other countries (e.g., Germany, Korea or Japan).⁵³

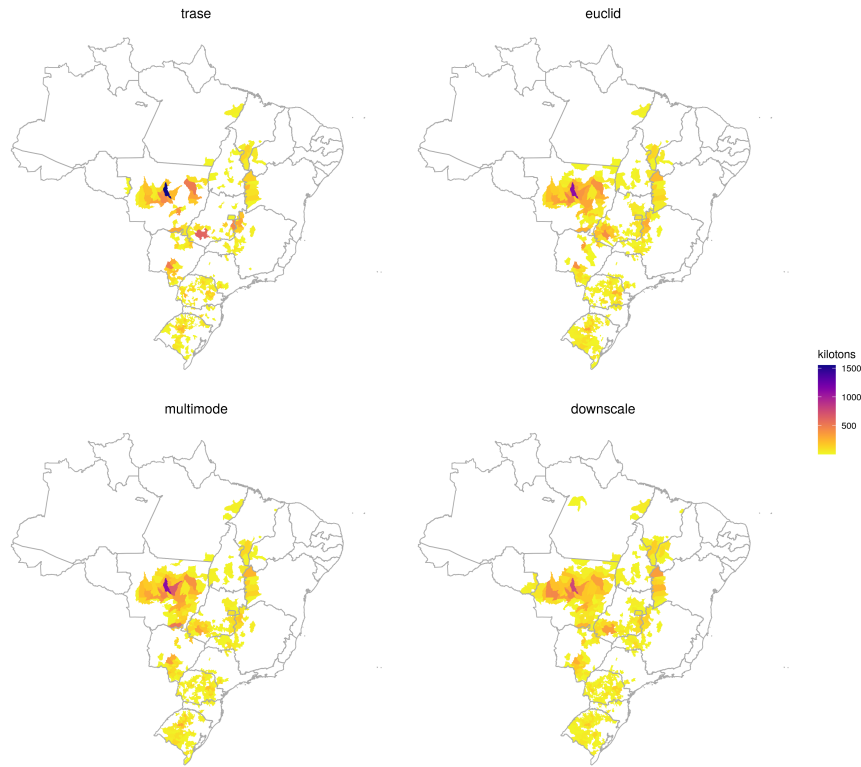


Figure 4: Source of soy exported to China across the four benchmarks

To assess the robustness of the above results, the same analysis is repeated using the root mean squared error (RMSE) instead of Pearson’s r as a comparison measure. In contrast to Pearson’s r , which is a relative measure of correlation, the RMSE is an absolute measure of the deviation between the two models in the same unit as the modeled flows. RMSE is sensitive to the range of the data and the absolute size of errors, i.e., it penalizes deviations in large flows more than deviations in small flows. It thus gives a complementary perspective on the models’ proximity to Trase. Table B3 in the appendix shows the corresponding results, again for a direct comparison as well as a comparison of focal means based on queen contiguity and 100 km radius neighborhoods. The results widely reflect the picture given by correlation analysis: When looking at individual importers, the subnational models generally perform better than the pure downscaling (i.e., they have a lower RMSE), and this

⁵²Spain has a particularly high dependence on Brazil as a source of soy, which is mainly consumed domestically and not re-exported (Hiel et al., 2020). Moreover, it imports most soy in the form of raw soybeans.

⁵³For additional map examples, see appendix C.2.

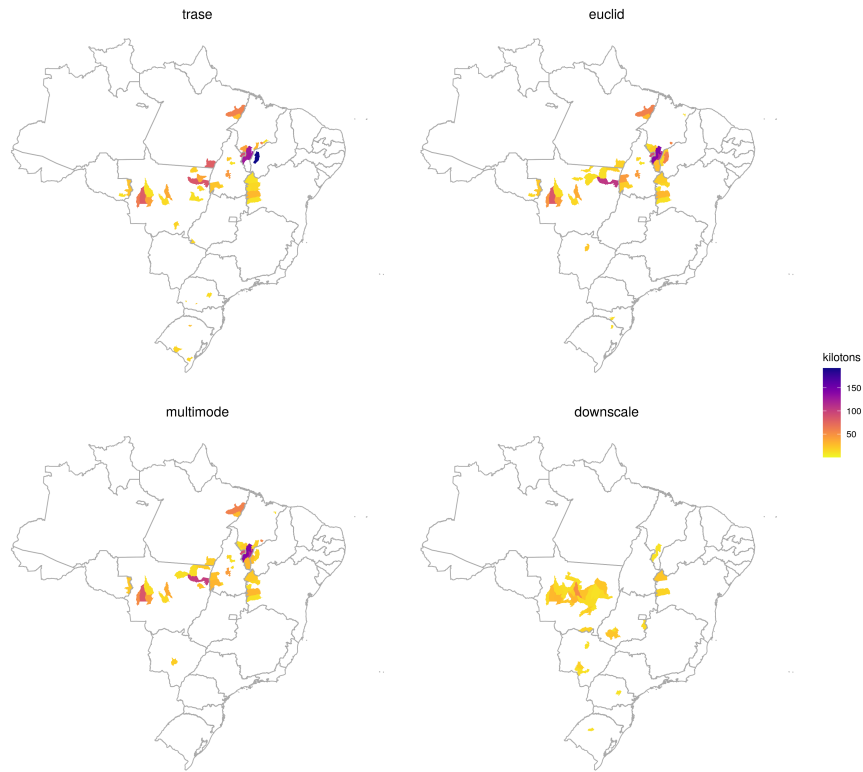


Figure 5: Source of soy exported to Spain across the four benchmarks

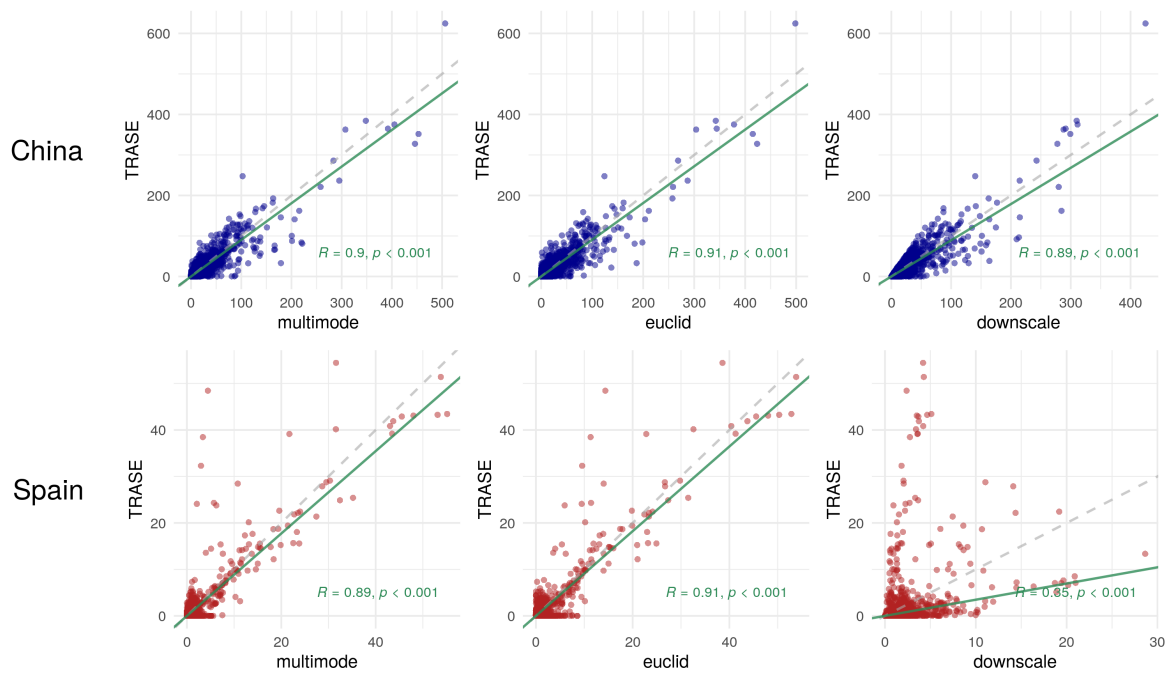


Figure 6: Scatterplots of focal model flows compared to Trase for China and Spain

performance difference becomes more pronounced as import volumes decrease. Likewise, the multi-modal model mostly gives slightly improved results compared to the Euclidean one. The benchmark results thus seem relatively robust to the chosen metric, with more elaborate comparisons left for future work.

4.4 Implications

The key findings of this comparative exercise can be summarized in three key points. First, when looking at all flows at once, the developed model framework (i.e., the Euclidean and multi-modal model) does not come closer to the results of Trase than a simple downscaling. This suggests a small overall value-added of the framework and also a small value-added of the multi-modal model compared to the Euclidean one. Second, however, the overall results seem to be driven by the flows to the main importers, especially China. For the majority of other importers, the subnational models generally perform better than downscaling, suggesting that the explicit modeling of subnational flows is an important aspect when investigating the footprints of individual nations. One might conclude that the smaller and more specific the investigated import flows, the more important a subnational model becomes. Third, in most cases, the multi-modal model performs only slightly better than the Euclidean one, suggesting that the inclusion of real-world features in a transport optimization model might not be strictly necessary.⁵⁴ The subnational estimation of domestic demand (i.e., the different use categories of the CBS) may have a larger influence on results than the transport setting, as suggested by the discrepancies in the flows remaining in Brazil.

To return to the two questions posed initially in this section, one can thus infer from the benchmarking that 1) the open-source model framework proposed here provides more meaningful results than the often practiced downscaling approximation and 2) the simple Euclidean distance model may suffice to approximate agents' transport decisions. Again, however, these first results need to be taken with caution in the light of the mentioned data discrepancies that prevent a fully "fair" comparison to Trase, and keeping in mind that the data of Trase is itself based on assumptions and uncertainties.

5 Discussion

This thesis proposed an open-source method for the modeling of subnational commodity flows applied to the case of Brazilian soy and conducted a benchmarking exercise to contextualize its results. This section reflects on the key implications drawn from the results, gives an outlook on extensions and future developments regarding the model framework, and discusses its limitations. Finally, it also briefly addresses the policy relevance of subnational supply chain data.

5.1 Advancing subnational supply chain modeling

The proposed framework and the comparative analysis can be seen as an attempt to gauge the sphere of subnational material flow analysis within its trade-offs between accuracy,

⁵⁴However, the impact of the transport distance or cost specification on model outcomes may require some additional considerations beyond the scope of this thesis.

openness and generalizability. The initial benchmarking results highlight the need for quality assessment in this field. This is all the more important as spatially explicit models are often employed to assess questions of environmental responsibility and supply chain governance (see also subsection 5.4 at the end of this chapter). The Trase initiative has taken a deliberate step here by extensively integrating company-specific data into its framework, allowing the so far most precise tracking of commodity flows. However, researchers may also want to model subnational flows of specific commodities in the absence of such data or the financial means to acquire them. Open-data and open-source models can play a role here, potentially spurring collaborative efforts to refine, extend and validate such models. This can include the integration of new public data sources, or potentially even inputs and micro-validations from individual supply chain actors.

The results of this thesis suggest two main takeaways. First, subnational modeling based on open data is preferable to pure downscaling, especially when investigating individual importing countries that have a distinct sourcing pattern (or specific downstream consuming countries via input-output linkages, see section 5.2.1). Second, a simple approach to transport optimization, i.e., Euclidean distance or simple road network distance minimization, can be a sufficient approximation, given that the model is founded on a solid subnational data basis that quantifies the range of subnational supply and use items with the best information possible.

Yet, results also underline that specific results generated by the model have to be interpreted with caution. Benchmarking showed that the accuracy (relative to Trase) can vary significantly on a country-to-country basis, and the sensitivity analysis of the transport model unveiled some (minor) sensitivities to input parameters. In this regard, it also becomes apparent how finespun and multi-faceted subnational supply chain models are, with several bottlenecks and elements of uncertainty. Further investigations into the drivers of this uncertainty may be valuable but are outside the scope of this thesis.

5.2 Extensions

The proposed model only represents a first element of an integrated toolbox for the spatially explicit assessment of telecoupled processes. To harvest its full potential, several further development steps are envisaged. Regarding the broader framework, the model is designed to be extended on both sides of the supply chain (the supply side and the use side). In particular, it is built to be fully compatible and integrable with FABIO to trace flows further downstream to the point of final consumption on the use side. On the supply side, both the refinement of production activities from the municipal to the grid level and the extension with related environmental pressures and impacts represent key steps for policy-relevant assessments. Concerning scope, generalizations of the framework towards a broader temporal, geographic and commodity coverage are planned. Below, I elaborate briefly on these points.

5.2.1 Linkage with FABIO

Within telecoupled systems, goods often pass multiple borders and transformation steps before being embodied into final consumer goods. In the case of soy, the majority of the commodity is transformed into soy cake, then fed to livestock, which is then slaughtered,

processed and prepared before it is consumed. All these tasks along the supply chain can potentially be carried out in different countries or regions. To trace commodities embedded in such sequences of intermediate exchange on a global scale, the use of MRIO analysis provides a powerful and prominent tool (Giljum et al., 2015, Minx et al., 2009, Wiedmann et al., 2006).

FABIO is a physical MRIO system based on official public statistics, comprising 130 agricultural and food products (including soybean, soy oil and soy cake) for 191 countries (Bruckner and Kuschnig, 2020). A hybrid version, integrating FABIO with EXIOBASE (Stadler et al., 2018) and thereby additionally covering more than 50 industrial sectors, was recently applied by Sun et al. (2022a,b). The subnational model framework proposed here was designed to be compatible and harmonized with FABIO in all its components, with the subnational supply and use database constructed in full accordance with the structure and base data of FABIO. This feature allows the subnational supply chain to be *nested* into the MRIO table to create a *spatially explicit input-output model* (Sun et al., 2018), as has been demonstrated with other subnational trade and IO data (Bachmann et al., 2015, Croft et al., 2018, Fry et al., 2022, Jiang et al., 2020, Towa et al., 2022). Notably, Croft et al. (2018) conducted a similar integration of Trase data with the MRIO IOTA and Green et al. (2019) used this framework to investigate consumption-based biodiversity impacts of soy cultivation in the Brazilian Cerrado.

The nesting procedure into FABIO involves the integration of the subnational data into all construction steps of the MRIO system, essentially replacing national sectors (e.g., soybean cultivation and processing in Brazil) with their subnational counterparts. This will be the topic of a separate study and will enable the tracing of subnational flows and related environmental impacts towards global final consumption at an unprecedented level of detail.

5.2.2 Linkage with gridded land use

On the production side, a refinement of production locations from the municipal to the grid level can be done with gridded land-use data. For Brazil, such a dataset is provided by the MapBiomass project (MapBiomass, 2021).⁵⁵ In the absence of sub-municipal information, this would be a simple downscaling exercise, albeit already from a high resolution. Nevertheless, it can help localize likely locations of land-use pressures within municipalities.

As pointed out by Moran et al. (2020), the increasing availability of high-resolution earth observation data and derived products such as agricultural production, water availability or land cover change maps open large and partly untapped potentials for spatially explicit supply chain assessments. Subnational economic data and models like the one presented here can provide an important connecting point at the interface between local economic and environmental processes.

5.2.3 Associated environmental pressures and impacts

Related to this, a key dimension of interest in material flow analysis is the associated local environmental consequences, i.e., the linkage of flows with their *footprints*. Such an environmental extension can include both *pressures* arising from local production and the consequent

⁵⁵MapBiomass provides land cover and land use maps in a resolution of 30 meters for the whole territory of Brazil. The data is based on Landsat images and openly available at <https://mapbiomas.org/>.

impacts. Pressures include changes in the environment such as material extraction, land and water use, nutrient leaching or greenhouse gas emissions, while impacts are related to undesirable changes in the environment, such as LUC, biodiversity loss, eutrophication or water scarcity.

Ideally, such information should be gathered at the highest spatial resolution possible. However, it also has to be pointed out that the relationship between pressures and impacts is highly complex, often involving indirect spatio-temporal dynamics such as spillovers, time lags or feedback effects (Parra Paitan and Verburg, 2019, Walker and Johnston, 1999). For example, and as mentioned above, the deforestation associated with soy cultivation is often the consequence of an indirect forest - cattle pasture - soy dynamic and a multi-faceted competition for land (Arima et al., 2011, Barona et al., 2010, Kuschnig et al., 2021, Song et al., 2021). Thus, questions about impacts need to look at the bigger picture by considering interactions between different socio-economic processes and actors.

5.2.4 Generalization: time series, products and regions

To get such a bigger picture, the extension of the model towards a broader temporal, thematic and geographic coverage can play a crucial role. Socio-ecological phenomena are inherently dynamic, requiring time series data. The case study in this thesis was conducted for the year 2013, but is easily generalized to other years, considering temporal changes in all underlying model parameters.

The framework is also flexible concerning the extension to other commodities and regions. Future enhancements in this direction are planned, contingent on data availability which needs to be assessed on a case-by-case basis. In this regard, the open-source nature of the framework can contribute to shared and collaborative progress, potentially even including collective data gathering, model improvement and validation.

5.3 Limitations

The presented approach relies on the imputation of missing data and several simplifying assumptions, giving rise to important limitations. Concerning the data basis for the case of Brazil, the model heavily relies on the municipal export data from Comex. As mentioned above, it assigns export flows to the municipalities where the respective trading company has its fiscal domicile. Consequently, the model assumes that the commodity flows are physically handled in or near these locations. This may not be appropriate in all cases, especially when these tax domiciles are located in commercial centers distant from agricultural production regions. As a major source of uncertainty, this calls for caution and further inquiries into the issue, potentially taking "unknown origin" cases in Trase as an orientation. Other elements of the subnational database are also associated with uncertainty (e.g., the feed and food use estimation), although special attention was paid to integrating diverse proxy information with a high level of spatial detail.

The case study has shown that the increasing public availability of economic and geospatial information in more fine-grained resolutions can provide a useful bedrock for the comprehensive subnational modeling of supply chains. However, as is the case for the municipal trade data used here, it is also evident that public data often masks the view to identify crucial individual supply chain actors such as trading companies. Research has highlighted

the significant role and transformative capabilities of such intermediate nodes in the supply chain, which themselves share responsibilities for environmental footprints together with producers and consumers (Godar et al., 2016, Lenzen et al., 2007, Sonnino et al., 2014). The open-source model presented here lacks the actor-centeredness of models like Trase. However, additional actor-specific information could in principle be flexibly integrated into the framework whenever available.

As regards the transport model, both the assumption of cost minimization as a key decision principle and some specific features of the model need to be critically evaluated. The multi-modal model assumes a certain range of possible transport configurations (see figure 2), linear costs and that speed limits can be used as a proxy for road quality. Moreover, not all of its parameters (e.g., the weighting of road distances by speed limits) are included in the sensitivity analysis, which should ideally encompass all exogenous factors of the model. Nevertheless, the multi-modal transport model is an attempt to integrate more real-world features into the allocation problem, something that is done rather rarely in this realm. Importantly, it showed that additional information (e.g., partial cargo flow data) can be flexibly integrated into such a model to improve its accuracy. Even though the benchmarking suggests that such a more complex setting might not be necessary (which is an insight in itself), it may be of value in cases where export data is not as detailed as in Brazil (e.g., only for ports or main trade hubs). On a technical note, it also has to be mentioned that the increased complexity of the multi-modal model comes at the cost of increased computational demands and the need for a commercial GAMS license.⁵⁶ Besides the model framework itself, the benchmarking exercise also exhibits some noteworthy limitations. The statistical comparison is rather coarse and based on simple correlation (and RMSE) analysis. Although the focal sum approach can account for spatial clustering to some degree, more targeted approaches may exist that perform better in the detection of spatial pattern similarities. Notably, the absence of systemic (and public) real-world validation data constrains the significance of all benchmarking results, relying on Trase as the key yardstick. Such data would be highly valuable both in "training" and validating subnational models.

5.4 What for? Material flows, responsibility and policy

To conclude the discussion, some final remarks on the general purpose and value of subnational material flow modeling may be appropriate. Models like the one presented here target a middle ground between national-level footprint analysis and highly specific life cycle assessments. Essentially, they try to "zoom in" on localized production and environmental effects of specific commodities while keeping the global scope of the production network they are embedded in. As such, they aim to provide a backbone for refined footprint analyses, informing policymakers, supply chain actors, academia and the general society on the fine-grained implications of consumption patterns and the consequent formulation of targeted policy interventions (Moran et al., 2020, Sun et al., 2018). As Moran et al. (2020, p.5) put it: "Spatially extended economic accounts are a part of the fundamental knowledge infrastructure necessary for maintaining a global economy that operates within planetary boundaries".

⁵⁶The number of variables in the model exceeds the limit of the free GAMS community license. For more information on GAMS licensing, see https://www.gams.com/latest/docs/UG_License.html. For details on the model implementation in GAMS, see appendix A.2.1.

At least three broad points can be emphasized regarding the application of such fine-scale assessments. First, and in the most general sense, the subnational disaggregation of supply chains allows for numerically more accurate importer- or consumption-based footprint assessments. Footprints on the national level rely on average impact factors per unit of production and therefore neglect that these impacts can vary significantly across subnational regions. This is especially relevant for large and heterogeneous countries such as Brazil, China or India. By tracing commodities to their subnational origins, the environmental footprints of different consumer regions, which are often subject to political and social discourse, can simply be calculated with higher precision. For MRIO-based calculations, it has been shown that the spatial or sectoral disaggregation of supply chains yields significantly higher accuracy in the derived environmental effects, even if this disaggregation is based on limited data ([de Koning et al., 2015](#), [Lenzen, 2011](#), [Su and Ang, 2014](#)). Furthermore, the approach can help in identifying supply chains that exhibit particularly strong concentration of footprints, pointing out the need for further care in increasing the level of detail regarding geographic and product resolution.

Second, a more precise representation of impacts associated with trade and consumption can serve as a risk analysis and decision support tool for downstream supply chain actors. Although the results presented here remain on the macro level, i.e. do not go down to the level of individual companies, they provide a graspable overview of the fine-scale origins of flows reaching a certain demand location, giving guidance to both industrial users and final consumers to make more informed procurement, investment and consumption decisions ([Fry et al., 2022](#), [Godar et al., 2016](#), [Stokeld et al., 2020](#)). The linkage to MRIO data can also reveal differences in footprint patterns resulting from the consumption of different downstream products (e.g., meat vs. biofuels) and highlight critical channels in global supply chains that may be less salient but entail significant spatially concentrated upstream risks.

Third, the effectiveness of conservation and trade policies crucially depends on the knowledge of existing socio-economic inter-linkages. The more precise this knowledge is, the better the understanding of the influence of different consumer countries or regions on critical environmental issues such as deforestation or biodiversity loss. This, in turn, provides fundamental information to align political strategies with the goals of multilateral conservation policies such as the New York Declaration on Forests ([United Nations, 2014](#)), the Amsterdam Declaration ([AD Partnership, 2015](#)) or the proposed EU regulation on deforestation-free products ([Bellfield et al., 2022](#), [European Commission, 2021](#)). From a regulatory perspective, it can assist both in monitoring the effectiveness of existing initiatives such as zero-deforestation commitments ([zu Ermgassen et al., 2020](#), [Trase, 2019](#)) and in the formulation of targeted policies with the appropriate geographical and topical coverage to minimize unintended consequences ([Greens/EFA, 2022](#)). For example, prominent policies such as the soy moratorium decreased deforestation in the Amazon but may have induced increased land-use change in the threatened Cerrado biome ([Moffette and Gibbs, 2021](#)), provoking calls to expand the geographical scope of the agreement ([Soterroni et al., 2019](#)). Withal, the visually concise messages transported by fine-scale footprints can serve as a useful lever for campaigns of civil society organizations and NGOs ([McNevin et al., 2020](#)). Ultimately, the improved traceability and transparency can contribute to efficient public-private policy mixes needed to achieve truly sustainable supply chains ([Lambin et al., 2018](#)). For both exporting and importing countries, it may thus be a valuable political resource.

However, here it is again worth emphasizing that the connection of material and land-use footprints – as highly resolved as they may be – with questions of responsibility is not clear-cut. As mentioned above, land-use decisions and their environmental effects are often provoked by a complex interplay with other subnational processes within the competition for limited resources and land. This competition may be caused by demand for the same or other commodities, driven by consumption in different foreign countries or the domestic market. Responsibility considerations and policy-directed analyses need to be aware of these interactions and all agents involved.

Thus, spatially explicit material flow models are only one part of the equation. Nevertheless, their part is of crucial importance, since the networks that link producers with consumers are becoming increasingly complex and fragmented, and rough approximations such as pure downscaling provide little help in disentangling this complexity. What the framework presented here ultimately intends, is to exploit the widest range of openly available data to shed light on the webs of subnational-to-international flows, and contribute to the open debate on the relevance and advancement of such models. The more reflected and well-founded the infrastructure of such models is, the more relevant and useful they become for the political debate.

6 Conclusion

Fine-scale material flow analyses beyond the abstract level of aggregate countries can improve the understanding of heterogeneous producer-to-consumer linkages and therefore enhance the value of macro-level tractability tools. Yet, the field of such "middle-ground" approaches emerged only recently, and cost-benefit considerations might steer scholars interested in the topic towards more simple but inaccurate downscaling approaches. Regarding this issue, the contribution of this work consists of two aspects: First, I presented a flexible framework for fine-scale supply chain modeling, distinguishing itself through its openness and compatibility with FABIO. With some effort, it can be reproduced, complemented with further data, or transferred to other commodities and regions. Second, regarding its applicability, I showed that such an open-source framework can yield significantly improved results compared to simple downscaling approximations, especially when investigating specific importer profiles. Issues of uncertainty and sensitivity were highlighted, suggesting that basic transport allocation rules can be a sufficient approach, but also urging caution in designing such models and interpreting results.

The focus of this thesis was largely methodological, leaving concrete applications of the framework to follow-up studies. Key insights can be gained when connecting the model with FABIO on the demand side and high-resolution environmental information on the supply side. This can unveil global telecoupled interactions at an unprecedented level of detail, and inform policymakers and diverse stakeholders regarding targeted measures to make supply chains more sustainable.

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A Methodological Appendix

A.1 Completing the subnational data basis

A.1.1 Geographical units: Brazil's municipalities

The municipalities (*municípios*) of Brazil are the country's lowest common level of geographical and political division. They subdivide the 26 Brazilian states (not including the Federal District of Brasilia, which is a separate federative unit beside the states). Since the last administrative reform in 2013, there are 5568 municipalities. Moreover, the Federal District of Brasilia, which is not subdivided into municipalities, and the archipelago of Fernando de Noronha, which represents a state district pertaining to the state of Pernambuco, are also included as statistical units within the IBGE's municipal database, thus yielding a total of 5570 units covering the country's entire territory (IBGE, 2019).⁵⁷ Each municipality has a capital city (*cidade*) and can be subdivided further into districts (*distritos*), subdistricts (*subdistritos*) and neighborhoods (*bairros*), but the level and extent of subdivision vary across municipalities.

All municipalities have a unique seven-digit geocode allowing their unambiguous identification. The first two digits of this code indicate the state (of which the first stands for one of the five statistical macroregions and the second for the state within the macroregion – in clockwise order), and the remaining five digits identify the municipality within the state, e.g., 5107925 for the municipality of Sorriso in the state of Mato Grosso (51), which belongs to the Center-West macroregion (5). Figure A1 illustrates this territorial division of the country into states and macroregions, including the respective IBGE codes. Throughout the compilation of the subnational database, all necessary data is linked to the municipalities via this unique code. This also includes the geographic polygon shapefiles of municipal boundaries from the IBGE (2015), which provide important information for the geospatial analysis in all parts of the modeling workflow. In all geospatial computations of this study, the Brazil Polyconic projection of the SIRGAS 2000 reference system (EPSG code 5880) is used.

Table A1 presents some descriptive statistics on the Brazilian municipalities, including all relevant supply and use items for the soy supply chain that are compiled or estimated in the steps described in section 3.1 and below. The availability of agricultural, economic and geographic data at the municipal level is relatively high for Brazil, which greatly benefits the estimation of required supply and use data.

In the mathematical notation in this appendix, the following subscripts are used to refer to data (either pre-existing or calculated) on different territorial levels:

- n stands for the national level (e.g., from the national CBS)
- r for the level of the five macro-regions
- s for the 26 states plus the Federal District of Brasilia, and

⁵⁷The official geographic data set defining the borders of these territorial units additionally includes two waterbodies as separate entities: The Lagoa (lagoon) Mirim and Lagoa dos Patos in the state of Rio Grande do Sul. Both cover only water surfaces and thus have no inhabitants or political status. However, they belong to the Brazilian mainland territory (without being part of any municipality) and are thus included in the data. In total, the dataset thus contains 5572 units.

Figure A1: Brazilian states by macroregion



Source: Author's visualization based on data from IBGE (2015) and auxiliary information from IBGE (2019).

- m for the 5570 municipal units

A wide range of data on the national, macroregional and state level, as well as gridded spatial data in different resolutions, is used in the estimation of municipality-level CBS items.

A.1.2 Imports and exports

The Comex database uses the 4-digit Harmonized Commodity Classification and Coding System (HS). Data for soybean (HS code 1201), soybean oil (1507) and soybean cake (2404) are contained as distinct commodities. Comex uses a slightly different state coding system compared to the IBGE, which required some corrections in order to match municipalities correctly.⁵⁸ I also harmonize the sample of origin countries (imports) and destination countries (exports) with the country sample of FABIO (which corresponds to the sample of the FAO's

⁵⁸São Paulo was changed from 34 in Comex to 35, Mato Grosso do Sul from 52 to 50, Goiás from 53 to 52 and the Federal District from 54 to 53.

Table A1: Descriptive statistics of CBS items across municipalities in 2013

	n	mean	sd	min	max	sum
General						
Population	5570	36.1	212.7	0.8	11821.9	201032.7
Area	5572	1545.5	5673.8	3.6	159549.5	8611779.2
Soybean						
Area planted	1963	14.2	36.5	0	619.9	27948.6
Production	1963	41.6	106.8	0	1926.3	81699.8
Imports	25	11.3	19.9	0	79	283
Exports	279	153.4	459.1	0	4038.6	42796.5
Processing	69	514.7	404.7	70.3	2674.3	35513.2
Feed	5493	0.1	0.4	0	15.5	637
Seed	1963	0.6	1.6	0	28.6	1211.5
Stock	1716	0.6	1	0	13.7	1099.9
Soy oil						
Production	69	102.6	80.7	14	532.9	7077
Imports	11	0.5	1.4	0	4.6	5.8
Exports	125	11	31	0	255	1380.6
Food	5570	0.5	2.9	0	159.5	2712.7
Other	5570	0.5	3.2	0	175.8	2989.6
Stock	69	0	0	0	0	0
Soy cake						
Production	69	400.3	314.8	54.7	2080	27622
Imports	5	2.1	3.5	0.1	8.2	10.6
Exports	178	74.9	198.3	0	1424.4	13333.5
Feed	5528	2.6	6.9	0	193.3	14299.1
Stock	69	0	0	0	0	0

Notes: Statistics refer to municipalities where the respective variable is unequal to zero. Population numbers in 1000 persons are from [IBGE \(2022c\)](#). Area per municipality in km² is calculated with polygons from [IBGE \(2015\)](#). Area planted refers to soy area (km²) from [IBGE \(2022a\)](#). All CBS variables are in kilotons. Sources and calculation methods of these variables are described in sections below. All values refer to 2013 and are balanced and harmonized with the national CBS values.

bilateral trade data). Countries that are not explicitly contained in FABIO are aggregated into a "Rest of the World" (ROW) region.⁵⁹

A.1.3 Soybean processing use and production of soybean oil and cake

Based on annual surveys of all oil extraction companies, [ABIOVE \(2013\)](#) publishes data on state-level processing capacity (in tonnes of daily oil crops processed) as well as a list of the individual surveyed facilities, including information on their location and the type of processed oil crop. As processing capacity by facility is not published, I distribute state-level capacity equally to all facilities within each state.

Next, I remove all facilities that only process crops other than soy, such as cottonseed, sunflower seed, or almonds. To obtain an estimate of the daily soy processing capacity by municipality, I sum the capacity of all active facilities within each municipality.⁶⁰ Finally, to get from daily processing capacity to an estimate of annual processing use, I multiply

⁵⁹The country sample of Comex is more detailed than the one of the FAO bilateral trade data, which is a fundamental source of FABIO. Aside from the aggregation of ROW countries, French Guiana, Guadeloupe, Martinique and Réunion are added to France, as they have been part of the French customs area since 1996 and are therefore treated as part of France in the FAO trade data ([FAO, 2003](#)).

⁶⁰The ABIOVE data also includes inactive facilities, which means that they did not operate in the given year. Besides processing capacity, the data also includes oil refining and bottling capacity. Both are not used in this

daily municipal capacities ($proc_cap_m$) by the ratio of annual national soybean processing use according to the CBS for Brazil ($proc_n^{bean}$) and the sum of daily processing capacities across all active facilities.⁶¹

$$proc_m^{bean} = proc_cap_m \frac{proc_n^{bean}}{\sum_m proc_cap_m} . \quad (A.1)$$

This yields the final estimate of municipal soybean processing use.

Having an estimate of soybean processing use per municipality, only one further step is required to obtain an estimate of municipal oil and cake *production*. By dividing the national CBS production quantities of these products ($prod_n^{oil}, prod_n^{cake}$) by the national soybean processing use, I obtain average conversion factors, which indicate how many tonnes of oil or cake are – on average – derived from one tonne of processed soybeans:

$$conv^{oil} = \frac{prod_n^{oil}}{proc_n^{bean}} = 0.199 \quad (A.2)$$

$$conv^{cake} = \frac{prod_n^{cake}}{proc_n^{bean}} = 0.778 . \quad (A.3)$$

Accordingly, for the year 2013, one tonne of crushed soybeans eventually delivered 0.199 tonnes of oil and 0.778 tonnes of cake on average. These conversion factors are consistent with values noted in the agrotechnical literature (Dei, 2011, Smaling et al., 2008).⁶² The remaining share can be interpreted as a loss factor, indicating how much biomass is lost in the process as waste:

$$conv^{loss} = 1 - conv^{oil} - conv^{cake} = 0.023 . \quad (A.4)$$

Municipal oil and cake production is then simply computed by multiplying municipal soybean processing use by the respective national conversion ratio:⁶³

$$prod_m^{oil} = proc_m^{bean} conv^{oil} \quad (A.5)$$

$$prod_m^{cake} = proc_m^{bean} conv^{cake} . \quad (A.6)$$

By taking the reciprocal of the sum of the conversion factors for oil and cake, one can obtain a mass-based commodity-equivalence factor (Trase, 2020):

$$equi_fact = \frac{1}{conv^{oil} + conv^{cake}} = \frac{1}{0.199 + 0.778} = 1.023 . \quad (A.7)$$

study, as most facilities that crush soybeans also refine (parts of) the produced crude oil, and not all crude oil is also refined

⁶¹This ratio can be interpreted as the average number of days per year that a processing facility is in operation and amounts to 234 days for the year 2013.

⁶²Smaling et al. (2008) report conversion ratios of 20% (oil) and 75% (cake), Dei (2011) states values of 19% and 79%.

⁶³This implicitly assumes that all processing facilities use the same technology.

This factor specifies how many units of raw soybeans are embodied in one unit of oil or cake and is useful for transforming flows of these products into equivalent flows of raw soybeans in the final stages of the model process.

A.1.4 Feed use

The feed use estimation takes municipal livestock numbers per specie from the IBGE (2022a) as base "population", but recognizes that the same specie might be fed differently, depending on local agricultural conditions. For this, the FAO's spatial GLW data and the GLEAM model (FAO, 2018), which is built on the former and its systems taxonomy, are used.

GLEAM is a process-based Life Cycle Assessment (LCA) model aiming to assess greenhouse gas emissions and commodity production of livestock sectors by production system on a spatially disaggregated scale. As such, it also provides system-specific feed rations for several production systems per specie and differentiated by world regions.⁶⁴ The included species and respective production systems are listed in table A2.⁶⁵

Table A2: Livestock management systems in GLEAM. Abbreviations in parentheses are used in calculations below.

Ruminants	Pigs	Poultry
Grazing (<i>gra</i>)	Industrial (<i>ind</i>)	Industrial (<i>ind</i>)
- meat (<i>meat</i>)	Intermediate (<i>int</i>)	- layers (<i>lay</i>)
- dairy (<i>dair</i>)	Backyard (<i>byd</i>)	- broilers (<i>bro</i>)
Mixed (<i>mix</i>)		Backyard (<i>byd</i>)
- meat (<i>meat</i>)		
- dairy (<i>dair</i>)		
Feedlot (<i>flot</i>)		

Note: Ruminants contain cattle, buffaloes, sheep and goats

Determining relevant livestock species

To first identify the species with significant soybean or soybean cake intake in Brazil, I consider both the GLEAM feed rations for the region of Latin America and the Caribbean and the literature on the soybean complex in Brazil. According to the Brazilian National Animal Feed Industry Union Sindirações (Zani, 2014), poultry, pigs and cattle accounted for 97% of soybean cake feed use in Brazil in 2013.⁶⁶ This is confirmed by studies on the Brazilian soy supply chain, which partly consider only poultry and pigs as main feed users (Caro et al., 2018, Kuepper et al., 2019, Smaling et al., 2008). In GLEAM, Pig and Poultry feed rations explicitly contain both soybeans and soybean cake for all production systems in the region. For Cattle and Buffaloes, soy is not explicitly stated as a feed item in GLEAM, but oilseed meals are included as an aggregate feed category. Furthermore, a study by Mottet et al. (2017) based on average feed rations for OCED and non-OECD countries from

⁶⁴Where available, the model uses country-specific feed rations (see FAO (2018, p.5)), but this data is not openly published by the FAO. The supplement data to the model only contains this information averaged over aggregated world regions. These are North America, Russian Federation, Western Europe, Eastern Europe, Near East and North Africa, East and Southeast Asia, Oceania, South Asia, Latin America and the Caribbean, and Sub-Saharan Africa. Brazil is thus subsumed in the region of Latin America and the Caribbean.

⁶⁵For a detailed description of all production systems, see FAO (2018, p.6).

⁶⁶Poultry has the highest share (62%), followed by pigs (24%) and cattle (11%).

GLEAM contains soybean cake as an explicit component of cattle and buffalo diets. For small ruminants (sheep and goats), soy is contained neither explicitly nor implicitly, both in the GLEAM documentation and [Mottet et al. \(2017\)](#). I thus consider poultry, pigs, cattle and buffaloes as relevant species to model municipal feed use. Buffaloes are included in order to avoid the underestimation of feed use in municipalities with a high concentration of Buffaloes, predominantly in the north of the country ([Bernardes, 2007](#), [IBGE, 2022b](#)).

Disaggregating species into production systems

Having identified relevant species, the municipal animal numbers from the IBGE Livestock Production Survey need to be disaggregated into the production systems used by GLEAM. For this, I use the GLW spatial database, which provides raster maps on the distribution of animal numbers across the world, including disaggregated data by production system in a resolution of 0.083333 decimal degrees (5 arc minutes or ~10 km at the equator). The GLW maps are also an integral data source in GLEAM, therefore their production system classification is largely congruent. The exceptions are that the GLW maps do not contain data on feedlot cattle, the distinction between dairy and non-dairy cattle (they are subsumed in the grazing and mixed systems), and the distinction between layers and broilers (they are subsumed under the *industrial* poultry system). I use auxiliary data from the IBGE to estimate these populations. The GLW rasters, which are only available for the year 2010, are used to compute the share of each production system in the total number of each species per municipality. These shares are then used to disaggregate the 2013 IBGE animal numbers.⁶⁷

For poultry⁶⁸ and pigs, the GLW data ([Gilbert et al., 2018a](#)) contains raster data representing the number of animals (in heads) of a given production system per pixel for the year 2010. For pigs, there are separate rasters for industrial (intensive), intermediate (semi-intensive) and backyard (extensive) systems, and the poultry rasters differentiate between industrial (including both layers and broilers) and backyard. I use these rasters in combination with the IBGE municipality polygons to obtain municipal shares of production systems by species. Therefore I first calculate the sum of animals of a given production system per municipality, using raster extraction. The method calculates the sum of all pixel values that fall within the boundaries of a municipality, weighted by the respective coverage fraction.⁶⁹ Consequently, the share of each production system in the overall number of animals per municipality is calculated, e.g., for industrial pig production (see table A2 for system abbreviations):

$$share_m^{pig_ind} = \frac{pig_ind_m^{2010}}{pig_ind_m^{2010} + pig_int_m^{2010} + pig_byd_m^{2010}} . \quad (A.8)$$

These shares are then applied to the IBGE animal numbers of 2013 to get the final number of animals per system:

⁶⁷By following this approach, I implicitly assume that the relative composition of production systems stayed constant from 2010 to 2013.

⁶⁸The GLW data for poultry covers only chickens and no other poultry birds such as turkeys. I assume that the production system distribution of chicken also applies to other poultry species.

⁶⁹This is conveniently done with the `exact_extract` function of the `exactextractr` package ([Bastion et al., 2022](#)). The applied summary statistic (in this case a simple sum) returns one value per polygon. To take into account that some pixels are only partly covered by a polygon, pixels values are multiplied by the respective coverage fraction before summation.

$$pig_ind_m^{2013} = pig_m^{2013} share_m^{pig_ind}. \quad (A.9)$$

To separate layers from broilers within the industrial poultry production system, I use data on the number of layer hens contained in the municipal livestock production data from the IBGE (2022b). Assuming that the layer numbers in the IBGE data concern both backyard and intensive systems, the disaggregation of the industrial system is computed as

$$poultry_ind_lay_m^{2013} = poultry_lay_m^{2013} share_m^{poultry_ind} \quad (A.10)$$

$$poultry_ind_bro_m^{2013} = (poultry_m^{2013} - poultry_lay_m^{2013}) share_m^{poultry_ind}. \quad (A.11)$$

For ruminant systems, the GLW data does not contain data with animal *numbers* per production system. Instead, it contains a single raster on *areas* of different production systems applying to all ruminant livestock (Robinson et al., 2018). This is because ruminants are "generally dependent directly on the environment in which they live for fodder and feed resources" (ibid.). The spatial dataset classifies global land area based on a typology of Seré and Steinfeld (1996) depending on origins of feed, stocking rates, the amount of irrigation and the length of the growing period. The classification distinguishes two main livestock systems: *livestock only* systems, representing systems where livestock husbandry based on local pastures and forages dominates other agricultural practices, and *mixed* systems, in which livestock husbandry co-exists with non-livestock farming activities and also relies on the by-products of these activities as feed.⁷⁰ Both of these main systems are further differentiated by agro-climate (*arid & semi-arid, humid & sub-humid, tropical highlands & temperate*) and mixed systems are also separated into *rainfed* and *irrigated* systems.⁷¹ Moreover, the GLW ruminant systems raster also includes classes for *urban areas, other tree-based systems* and *unsuitable areas* (water & snow/ice). To harmonize the classes in the raster with the ruminant systems present in GLEAM (grazing and mixed), I assume that all livestock only areas (predominantly) host grazing systems and all mixed areas host mixed systems. Moreover, I assign other tree-based and unsuitable areas to the grazing system⁷² and urban areas to the mixed system. Then, I combine the raster of ruminant system areas (now containing only the two aggregate classes grazing and mixed) with the GLW distribution maps of cattle (Gilbert et al., 2018c) and buffaloes (Gilbert et al., 2018b) (in heads per pixel) to obtain distributions of both species by production system. I then follow the same procedure as described above for poultry and pigs to derive municipal animal numbers per system for cattle and buffaloes in 2013.

To fully harmonize the ruminant animal number with the classification of the used GLEAM feed ratios, both cattle and buffalo numbers need to be split into dairy and meat

⁷⁰Robinson et al. (2011, p.6) give the following precise definitions: "Solely livestock systems are those in which more than 90% of dry matter fed to animals comes from rangelands, pastures, annual forages and purchased feeds, and less than 10% of the total value of production comes from non-livestock farming activities. Mixed farming systems are those in which either more than 10% of the dry matter fed to animals comes from crop by-products or stubble, or more than 10% of the total value of production comes from non-livestock farming activities."

⁷¹The original classification of Seré and Steinfeld (1996) also distinguished livestock-only systems into grassland and landless systems. However, the landless system class is not present in the GLW dataset.

⁷²This is mainly relevant as areas that were deforested between 2010 and 2013 might have become pasture areas.

animals, and for cattle, the feedlot system needs to be separated as well. To obtain numbers of feedlot cattle by municipality, I use data from the IBGE 2006 Agricultural Census (IBGE, 2006)⁷³ and extrapolate the numbers to 2013 using a growth rate in the national feedlot cattle herd taken from the Brazilian Beef Exporters Association ABIEC (2020)⁷⁴. To divide the cattle numbers into meat and dairy cattle, I use data on dairy cows from the 2013 Municipal Livestock Production Survey (IBGE, 2022b) and assume that the remainder of the municipal cattle herds represents meat cattle, of which a part is held in feedlots. Accordingly, the final numbers of cattle, e.g., in the grazing system, are obtained as

$$cattle_gra_dair_m^{2013} = cattle_dair_m^{2013} share_m^{cattle_gra} \quad (A.12)$$

$$cattle_gra_meat_m^{2013} = (cattle_m^{2013} - cattle_dair_m^{2013} - cattle_flot_m^{2013}) share_m^{cattle_gra} . \quad (A.13)$$

For buffaloes, the same procedure is followed (omitting the feedlot part as it is only relevant for cattle). As no data on dairy buffalo numbers is available, I assume that the share of dairy cows in the overall cattle herd is also representative of buffaloes.

Applying system-specific feed rations

The system-specific feed rations in GLEAM are indicated in percentages of total dry matter intake. Table A3 details the feed rations used in this study. For poultry and pigs, rations are taken from the GLEAM documentation (FAO, 2018) for Latin America and the Caribbean, while for cattle and buffaloes, data from Mottet et al. (2017) for non-OECD countries is used. Data on annual dry matter intake in kilograms per animal (DM_intake) is also obtained from Mottet et al. (values for non-OECD countries).

Using a dry matter content ($DM_content$) of 88% both for soybeans and soybean cake, as indicated by de Brum et al. (2006) as typical values for Brazilian livestock feed, the required annual amount of soybean feed per animal head in a specific system – e.g., industrial pigs – is given by

$$feed_head_{pig_ind}^{bean} = DM_intake_{pig_ind}^{bean} * ration_{pig_ind}^{bean} * \frac{1}{DM_content^{bean}} . \quad (A.14)$$

Finally, total feed use per municipality and species is then simply computed by multiplying the annual feed requirements per animal with the livestock numbers derived above, e.g., for the example of industrial pigs

$$feed_{pig_ind}^{bean} = feed_head_{pig_ind}^{bean} * pig_ind_m^{2013} . \quad (A.15)$$

Summing over all management systems then yields the total feed use of soybean and soybean cake on the municipal level.

Although I apply the feed use estimation described here only to Brazil, note that in principle it could be applied to any country or region worldwide. Even in the absence of official subnational data on livestock populations – which is used to get accurate municipal animal

⁷³The census is done only every 10 years.

⁷⁴According to this report, the total number of feedlot cattle in Brazil increased from 3.05 million in 2010 to 4.38 million in 2013, representing a growth rate of 27%.

Table A3: System-specific feed rations used in this study

Specie	System	Subsystem	DM intake (kg/year)	Soybean (%)	Soy Cake (%)
Cattle	grazing	meat	2680	0	0,15
		dairy	2680	0	0,15
	mixed	meat	2531	0	0,51
		dairy	2531	0	0,51
	feedlot	meat	1784	0	2,47
Buffalo	grazing	meat	2680	0	0,15
		dairy	2680	0	0,15
	mixed	meat	2531	0	0,51
		dairy	2531	0	0,51
Chicken	backyard	-	20	1,7	20,9
	layer	-	26	3,7	14,8
	broiler	-	34	0,0	27,0
Pig	backyard	-	506	1,5	18,0
	intermediate	-	563	3,6	28,2
	industrial	-	628	3,9	21,0

Sources: Dry matter intake from Mottet et al. (2017) representing average values for non-OECD countries. Feed ratios for chicken and pigs from FAO (2018), and for cattle and buffaloes from Mottet et al. (2017).

numbers for 2013 in Brazil – the spatial livestock distribution data from FAO would allow such an analysis on any level of regional detail (using an appropriate temporal extrapolation if needed).

A.1.5 Food use

The IBGE's Consumer Expenditure Survey (IBGE, 2018) contains data on annual per-capita purchases of a wide range of food items (in kilograms) differentiated by state. Soybean oil is reported as an explicit item, revealing significant regional differences (ranging from 2.5 kg in Paraíba to 7.2 kg in Goiás).⁷⁵ As soybeans are not contained as a separate item due to their low culinary relevance, I assume that their consumption follows a similar regional pattern as the consumption of soybean oil.⁷⁶

Thus, I allocate the national food use of both soybean oil and soybeans from the CBS to individual municipalities by using municipal population numbers pop_m , weighted by the state-specific per-capita soybean oil purchases $purch_{m,s}^{oil}$, e.g.

$$food_m^{oil} = \frac{pop_m purch_{m,s}^{oil}}{\sum_m (pop_m purch_{m,s}^{oil})} food_n^{oil}. \quad (A.16)$$

⁷⁵Note that the range of soybean oil purchases per capita from the IBGE data is well below the average soybean oil consumption per capita as computed from the CBS and the total Brazilian population in 2013 (2.71 Mt total oil consumption / 201.04 million inhabitants = 13.5 kg per capita). This can be explained by the definition of "purchases" in the Expenditure Survey, which includes only the direct acquisition of soybean oil (e.g., in supermarkets), but not the indirect acquisition via other processed foods or in restaurants.

⁷⁶As noted in the main text, the "Soybean" class in FAOSTAT's CBS statistics also contains soy sauce, soy paste and soy curd. These products play a minor role in Brazilian diets and are hence assumed to be consumed proportionally to soybeans.

A.1.6 Other use

The ANP provides a comprehensive range of statistical data on biofuel production in Brazil (ANP, 2022). For the year 2013, monthly bulletin statistics are used, including information on all active biodiesel facilities, their capacity and municipality (ANP, 2013). In December 2013, there were 64 facilities in operation, located mainly in the central and southern regions of the country, with a total daily production capacity of roughly 22 000 m³. Additionally, the data contains regional averages of the feedstock composition across the five macro-regions (with input shares of soy ranging from 32.15% in the South-East to 81.37% in the Center-West region), which I use to weight local production capacities. Municipal soybean oil use in the biodiesel sector is then estimated by allocating the national CBS value to each municipality according to its share in the total national feedstock-weighted biodiesel capacity:

$$other_m^{oil} = \frac{cap_m^{diesel} inp_share_{m,r}^{soy}}{\sum_m (cap_m^{diesel} inp_share_{m,r}^{soy})} other_n^{oil} . \quad (A.17)$$

A.1.7 Seed use

Deriving the seed use share on the national level from the CBS as

$$seedshare = \frac{seed_n^{bean}}{prod_n^{bean}} \quad (A.18)$$

and assuming that this share applies similarly to all regions of the country to retain a part of the production as seed material, municipal seed use is estimated via municipal production as

$$seed_m^{bean} = prod_m^{bean} seedshare . \quad (A.19)$$

A.1.8 Stock addition

The grain storage layer from the IBGE's transport logistics database (IBGE, 2014) gives the point locations of 1724 storage units, including their storage capacity.

Assuming that the storage activity (i.e., the composition of stored grains) is similar across the country, I use the aggregate storage capacity per municipality ($stock_cap_m$) derived from the data to spatially allocate the national CBS value:⁷⁷

$$stock_m^{bean} = \frac{stock_cap_m}{\sum_m stock_cap_m} stock_n^{bean} . \quad (A.20)$$

⁷⁷The IBGE grain storage data does not give information on the types of grains that are (preferably) stored in each unit. While this might mask regional differences in the storage structure, I assume that it still sufficiently proxies the storage demand for soy, given the low volume of stock additions relative to total production.

A.1.9 Final harmonization and balancing

The items that were estimated based on proxy information (blue cells in table 2) already match national aggregates by construction. For the soybean production and the trade of all three soy items, discrepancies with the national CBS exist (see table A4). These are minor in all cases, except for the import of soybean cake, which is considerably higher in the CBS, although on a low absolute level. The municipal data is simply harmonized with the CBS by re-scaling it proportionally according to the ratios displayed in table A4.⁷⁸

Table A4: Discrepancies between national CBS values and municipality-level data (kilotons).

	item	CBS	municipal	ratio
Soybean	production	81699.8	81724.5	0.9997
	imports	283.0	282.8	1.0006
	exports	42796.5	42796.1	1.0000
Soy oil	imports	5.8	5.0	1.1563
	exports	1380.6	1362.1	1.0135
Soy cake	imports	10.6	3.9	2.7405
	exports	13333.5	13333.5	1.0000

Notes: Municipal production data refers to [IBGE \(2022a\)](#) and trade data to [Comex Stat \(2022a\)](#).

Lastly, I adjust small pre-existing imbalances between total supply and use in the national CBS⁷⁹ via the stock addition item. For soybeans, this is done via a proportional reduction of stock addition in all municipalities with storage units, and for oil and cake, I allocate the low stock variations proportionally to the municipalities' oil and cake production.

A.2 Modeling subnational flows

A.2.1 Implementation of the multi-modal transport model in GAMS

The multi-modal model is implemented in GAMS as a linear model, with sets, parameters, variables and equations as described in section 3.2.2.⁸⁰ GAMS allows the efficient solving of large-scale optimization problems with numerous constraints. However, it has to be pointed out that even for GAMS, the described multi-modal model represents a very large model, as it requires the allocation of flows between 5570 municipalities across three transport modes for three products. In order to keep the model solvable, I make use of the so-called domain restriction functionality in GAMS (using the \$-operator for conditional expressions), which allows the restriction of equations to relevant pairs of origin/destination nodes, and therefore considerably reduces the size of the problem ([McCarl et al., 2014](#), [Rosenthal, 2017](#)). More precisely, it allows the domains of individual equation terms to be limited to i) municipalities with a positive excess supply (excess use) as a supplier (user) of a product and ii) rail and water hubs with a positive outflow (inflow) of a product according to the official cargo

⁷⁸Although the imports of soybean cake are thus more than doubled, I argue that this is still reasonable given their low absolute value compared to domestic production volumes.

⁷⁹For soybeans, there is an excess use of 126 t, for soybean oil an excess supply of 1 t, and for soybean cake an excess use of 5 t.

⁸⁰The GAMS code is available on <https://github.com/strsek/soyprint> alongside the R code.

data as origin (destination) hubs.⁸¹ This avoids unnecessary terms in the objective and constraint equations. The model is then solved using the CPLEX optimizer. Although using GAMS implies a departure from R, in which all other working steps are conducted, it can be seamlessly integrated into the R workflow with the interface provided by the `gdxrrw` package (Ferris et al., 2021). It allows the transfer of input and output data in GAMS format (`gdx`) and the calling of the GAMS script from within R.

A.2.2 Mode-specific transport cost

Road distance

For road transport, I use road network data from OpenStreetMap (OSM), taken from the country-level extract provided by Geofabrik (2018). To account for the fact that the year of analysis is 2013, I use historic data from 2014. Compared to other openly available road network data for Brazil, OSM is more extensive and detailed, minimizing the number of municipalities that are not connected to the network. For example, the dataset from the IBGE (2014) only contains federal roads, leaving 189 municipalities unconnected to the road network, while the OSM dataset only has 101 unconnected municipalities. Moreover, OSM has global coverage, facilitating the potential application of this method to other regions. To be able to approximate road travel distances to/from unconnected municipalities, I use a raster-based method, which simply assumes that areas without roads are "hard" to traverse, as expressed by a measure of *conductance*. This can be achieved in R with the `gdistance` package (van Etten, 2017), which enables the calculation of shortest paths between any two points on a raster surface by interpreting each pixel as a node of a weighted network and calculating shortest paths based on Dijkstra's algorithm (Dijkstra, 1959). The network is rasterized with a resolution of 5 km in order to keep computations at a reasonable level of efficiency.

Furthermore, it is well known that road quality can vary strongly in Brazil, which can significantly influence transport costs (Fliehr, 2013, Oliveira et al., 2016, Salin, 2021). To consider road quality in the computation of shortest paths and corresponding costs, I exploit attribute information on driving speed limits of roads from OSM as a proxy for road quality.⁸² When rasterizing the road network, these speed limits are then transformed into a conductance value indicating how easy it is to cross a pixel. For this, I roughly follow the estimates of CNT (2021) on operational cost increases relative to road conditions stated in the first column of table A5, and assume that different speed limits indicated in the second column loosely correspond to these conditions. The speed limit is then transformed into a conductance value according to

⁸¹For example, this means that municipalities that have zero excess supply of soybeans do not enter the objective function and constraints as a supplier of this product, or that rail stations that did not receive any soybean oil according to the data do not enter the equations as potential recipients of this product. I.e., the sets are selectively filtered as they enter the equations.

⁸²Although there is no harmonized method for setting speed limits in Brazil, they are generally set under consideration of road quality, e.g., surface type and condition (Contesini, 2013). Therefore I assume that the *maxspeed* attribute of OSM roads qualifies as a proxy for road quality. As not all roads on OSM have a *maxspeed* value assigned, I interpolate missing values by taking the value of the closest road of the same type. The road types (tags) considered are *motorway*, *motorway_link*, *primary*, *primary_link*, *secondary*, *secondary_link*, *tertiary*, *tertiary_link*, *trunk*, *trunk_link*, *service* and *track*.

$$conductance = \sqrt{\frac{maxspeed}{80}}, \quad (A.21)$$

where maxspeed is the highest speed limit among all roads falling into the area of a given 5 km × 5 km pixel. I thus assume that 80 km/h represents a road of regular quality. Areas without roads are assumed to imply a very low speed limit of 10 km/h and consequently a low conductance (column 3 of table A5). The reciprocal of the conductance values can be understood as the difficulty or *friction* of traversing a pixel or simply as a weight that decreases or increases the effective travel distance (column 4). More precisely, the distance to move between adjacent pixels is weighted by the mean of their friction values.⁸³ As a comparison, the last column of table A5 indicates how this friction translates into a cost markup relative to "very good" roads (120 km/h). Compared to the cost markups given CNT in the first column, the resulting markups are more conservative, giving less "punishment" to roads of lower quality.

Table A5: Weighting of road distances according to speed limits

	cost markup (%) (CNT, 2021)	speed limit (km/h)	conductance	friction	resulting markup (%)
Very good	0	120	1.22	0.82	0
Good	19	100	1.12	0.89	9
Regular	41	80	1	1	22
Bad	66	60	0.87	1.15	41
Poor	92	40	0.71	1.41	73
no road	-	10	0.35	2.83	246

Notes: *Seed limit* is the assumed speed limit corresponding to the road condition categories of CNT (2021). *Conductance* is obtained via equation A.21 and *friction* is the inverse of it. The last column gives the resulting cost markup based on these computations.

With each pixel having a maxspeed and corresponding conductance value, the shortest path between each pair of the 5570 municipalities is computed by minimizing the cumulative sum of friction costs between them.⁸⁴ As reference coordinates of each municipality, I use the coordinates of their capital taken from the IBGE (2010), assuming that they generally represent the economic and population centers of municipalities. The resulting $m \times m$ distance matrix of shortest paths $\mathbf{D}_{ab}$ with elements d_{ab} then gives, for each pair of municipalities, the network distance weighted by the friction cost of each traversed pixel. Likewise, the distance matrix $\mathbf{D}_{af}$ gives distances between each municipality capital and each origin rail/water hub, and $\mathbf{D}_{gb}$ the distance from the destination rail/water hubs to each municipality capital. Figure A2 shows an exemplary shortest path between Sorriso in Mato Grosso (one of the centers of soybean production) and the port city of Santos. The calculated (weighted) path distance amounts to 1910 km, while the corresponding fastest current route on google maps (Google, 2022) has a distance of 2007 km.

⁸³E.g., if two adjacent pixels (with a resolution of 5 km) both have a speed limit of 80, the effective distance between them is $5 \times \frac{1+1}{2} = 5$ km. If one of the two would have a speed limit of 120, the effective distance would be $5 \times \frac{1+0.82}{2} = 4.55$ km.

⁸⁴For this, the functions of the gdistance package also take into account that diagonal moves on the grid imply a larger distance than orthogonal moves, and if applicable also distortions stemming from equirectangular (lat-lon) grids (van Etten, 2017).

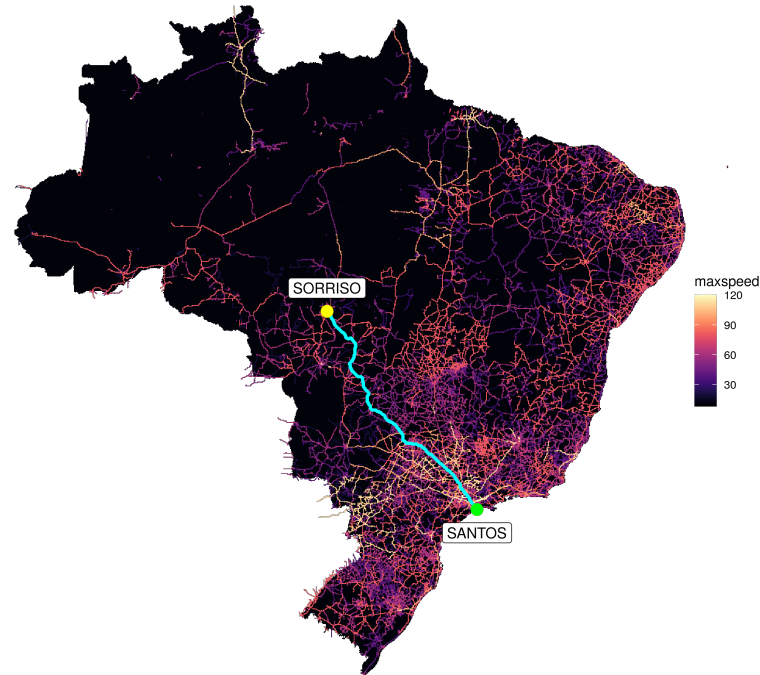


Figure A2: Exemplary shortest path on the road network

Rail and water distance

Railway network geodata is retrieved from the national land transport agency [ANTT \(2020\)](#), including all operational rail lines and stations for the year 2016.⁸⁵ Origin and destination stations for soy transport are filtered from the full set of stations using the officially reported rail cargo data published by ([ANTT, 2021](#)). These cargo records indicate origin and destination stations as well as the respective freight volume of each flow. Geographic data of the navigable inland waterways based on the national transport plan of 2014 is available from the [DNIT \(2021\)](#), and detailed data on the position of inland port terminals is obtained from [ANTAQ \(2020\)](#)⁸⁶. Analogous to the rail stations, the relevant origin and destination ports are filtered based on official water cargo records from [ANTAQ \(2021\)](#). In contrast to road transport, all rail stations and ports are located along the respective network.⁸⁷ Moreover, it is assumed that the conditions of operational rail tracks and navigable waters do not vary as much as for roads and hence do not influence transport costs significantly. Although the absence of non-connected nodes would allow a vector-based distance calculation for railways and waterways, I again use the raster-based method for reasons of consistency. As no quality-related weighting is required, the conductance of all pixels crossed by railway/waterway lines is equal to one, and the resulting shortest-path distances are unweighted

⁸⁵The data represents the entire federal railway mesh granted by the government to various railway operators. The rail stations also include stations operated by private (industrial) companies along the lines, such as soybean processing plants. The data is published annually by law, but shapefiles are only available since 2016.

⁸⁶Port geodata is also available from the Ministry of Infrastructure and the IBGE, but the ANTAQ dataset represents the most detailed one, including both public ports and private terminals.

⁸⁷This was checked by creating a buffer of 100 m around rail lines and 3000 m around waterway lines, and testing if all rail stations/ports fall within these buffers.

grid-network distances.⁸⁸ The results are two distance matrices, one for each pair of origin/destination rail stations and one for each pair of origin/destination ports, which can be subsumed in a single matrix with elements d_{fg} .⁸⁹

From distance to cost

To transform the calculated distances d into costs per tonne of hauled cargo c , I multiply distances with mode-specific cost values per tonne-kilometer c^{tkm} taken from the literature. E.g., for the road transport cost between each pair of municipalities, we get⁹⁰

$$c_{ab} = d_{ab} c_{\text{road}}^{\text{tkm}}, \quad (\text{A.22})$$

or for the rail and ship costs between each pair of rail stations and inland ports

$$c_{fg} = \begin{cases} d_{fg} c_{\text{rail}}^{\text{tkm}}, & \text{if } f \text{ and } g \text{ are rail stations} \\ d_{fg} c_{\text{water}}^{\text{tkm}}, & \text{if } f \text{ and } g \text{ are inland ports} \\ \infty & \text{otherwise.} \end{cases}$$

However, the cost values c^{tkm} found in the literature exhibit considerable variation, both in absolute terms and concerning the relation between transport modes. Thus, instead of relying on one specific cost value per mode, I define ranges based on a literature review. Table A6 summarizes the result of this review and the resulting ranges (smallest to largest value) per mode, expressed in 2013 US\$. The table also contains a column for transshipment (switching) cost which are added to the road costs in c_{af} and c_{gb} . The next section of the appendix describes how these cost ranges enter the optimization model.

A.2.3 Sensitivity analysis

To assess how sensitive the outcome of the model is to the exogenous cost parameters, I perform iterative Monte Carlo simulations, randomly picking from uniform cost distributions defined by the ranges in the bottom row of table A6. Such simulations can be useful to detect the influence of exogenous parameters and to obtain a range of uncertainty regarding the model outcome (Mavrotas et al., 2010). The model is run 1000 times, yielding 1000 different solutions.⁹¹

To compare the variability of flows across the different simulations, I compute summary statistics for each subnational origin-destination flow aggregated across transport modes (i.e., each element in $\mathbf{X}_{ab}^p$). Figure A3 compares the mean of each element to its coefficient of variation (CV), which is defined as the ratio of the standard deviation to the mean. The

⁸⁸Pixels crossed by the rail/waterway network obtain a conductance of one, and all other pixels obtain an "NA"-value, indicating that they cannot be traversed.

⁸⁹Routes between nodes of a different mode (e.g., between a rail station and port) are impossible, and thus the corresponding distance is infinite. The code keeps the distance matrices of rail and water separate, but they are combined here for rotational convenience.

⁹⁰Note that this procedure involves the implicit assumption of linear transport cost. Although costs generally follow a non-linear path due to the presence of fixed cost, I assume that their consideration would have only small effects on the outcome of the transport model.

⁹¹To increase efficiency, the iterations are run in parallel, using 12 cores and 135 GB RAM. Total execution time amounted to roughly 10 hours.

Table A6: Mode-specific cost values from the literature (US\$ per tkm)

	road	rail	water	switch
Caixeta Filho (2003)	0,031 - 0,04	0,022	0,017	
Caixeta Filho (2008)	0,038 - 0,058	0,015 - 0,023	0,012 - 0,018	
Vera-Diaz et al. (2009)	0,058 - 0,174			
Rebelo et al. (2012)	0,047	0,019	0,012	
Ripoll (2010)	0,054 - 0,102	0,053	0,031 - 0,032	2,573 - 2,573
Fliehr (2013)	0,047 - 0,057			
Monteiro et al. (2011)	0,055	0,02		
Reis et al. (2015)	0,048	0,026	0,004	
Antunes et al. (2015)	0,039	0,011	0,006	
ILOS (2016)	0,121	0,022	0,023	
Lopes (2017)	0,054	0,014 - 0,029	0,009	0,736 - 0,736
Brugnera and Dalchiavon (2017)	0,013 - 0,017	0,006 - 0,01		
Góes and Lopes (2018)	0,056	0,021 - 0,027	0,009	0,762 - 0,762
Salin (2021)	0,055 - 0,069			
Sifreca (2021)	0,032 - 0,063			
Range	0,013 - 0,174	0,006 - 0,065	0,004 - 0,032	0,736 - 2,573

Notes: All original values were transformed into 2013 US\$ using consumer price indices and annual exchange rates from FAOSTAT (2022b,c). Ranges in the bottom line represent the minimum and maximum value per column.

CV gives an indication of the variability of individual flows across simulations relative to their average size. The plot excludes flows with a mean of zero (i.e., which are zero in all simulations) but contains a large number of small flows compared to fewer large flows. The density curves along the axes indicate smoothed kernel densities of both variables. It becomes evident that the large majority of flows are rather stable across simulations, indicated by a low CV. Importantly, outliers with high volatility predominantly affect small flows (mostly soybean), while larger flows depict a much more stable pattern. To get a clearer picture of the influence of the more volatile but low flows on the aggregate outcome, figure A4 plots the CV against the cumulative sum of flows (means) ordered by their CV. This confirms that the volatile flows have a rather low weight in the aggregate sum of flows. Roughly 75% of the total soybean flow volume has a CV of less than one, and only a negligible share has a CV larger than five. For oil and cake, this pattern is even more pronounced. One can thus infer that the model is relatively insensitive to the mode-specific cost parameters. This is likely due to the specific nature of model constraints, especially those constraining the rail and water cargo volumes.

While figures A3 and A4 depict the sensitivity of flows on the subnational level, a similar analysis can be done after linking these flows to the importing countries and tracing oil/cake flows back to the primary soybean origin, as is done in section 3.3 in the matrix $Q_{a,k}^{bean}$. The corresponding plots (figure A5 and A6) show a similar picture compared to the subnational flows above. Outliers with high variability are even less prevalent here, as export flows are diversified across destination countries.

For the benchmarking exercises conducted in section 4, the average of subnational flows across all simulations is used as the core representative of the multi-modal network model.

A more detailed exploration of the model sensitivity could investigate the influence of each cost parameter and the resulting modal choice more closely, or look at the relationship between volatility and travel distance (i.e., whether volatility is more present among short-

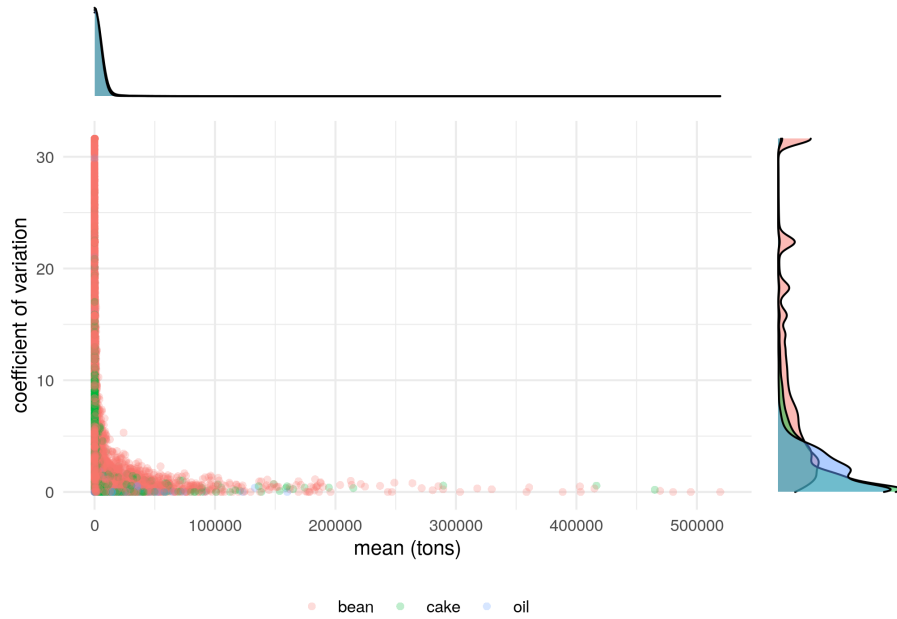


Figure A3: Variability of flows between municipalities across simulations. Each dot represents a flow between a specific pair of origin and destination municipalities. Density plots give the kernel densities of the respective axis, with a bandwidth of 5000 for the x-axis (mean) and 0.5 for the y-axis (CV). Only flows with a non-zero mean are included.

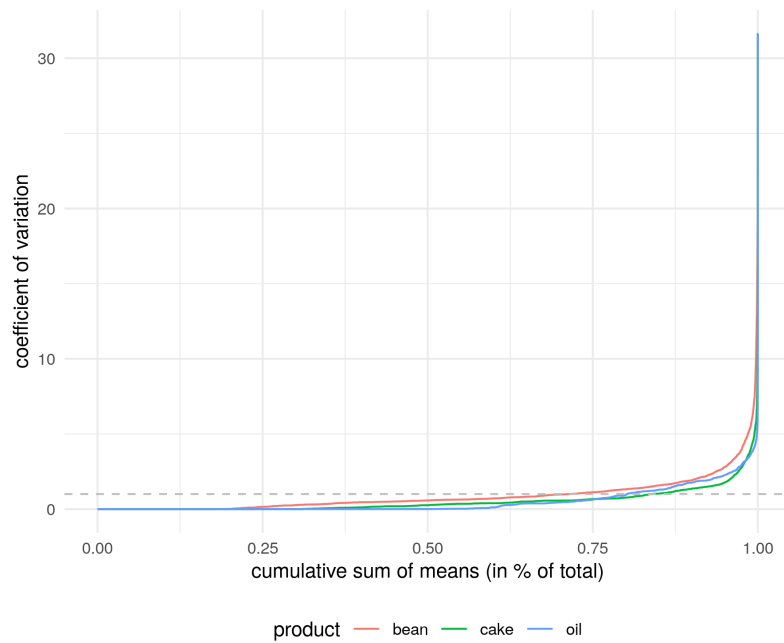


Figure A4: Variability compared to the cumulative sum of flows ordered by CV. The dotted horizontal line corresponds to a value of 1.

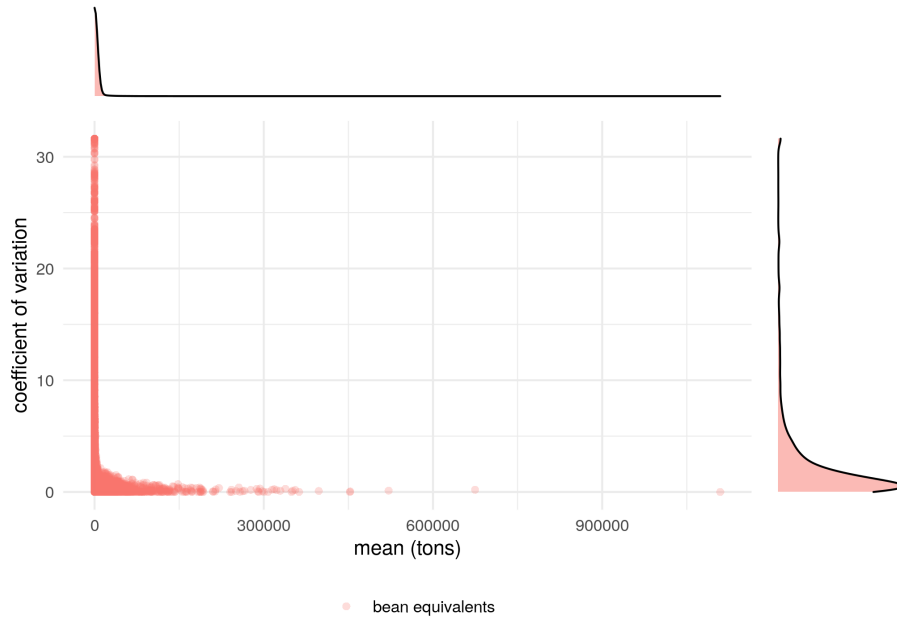


Figure A5: Variability of flows from origin municipalities to importing countries across simulations. Each dot represents a flow between a specific pair of origin and destination municipalities. Density plots give the kernel densities of the respective axis, with a band-width of 5000 for the x-axis (mean) and 0.5 for the y-axis (CV). Only flows with a non-zero mean are included.

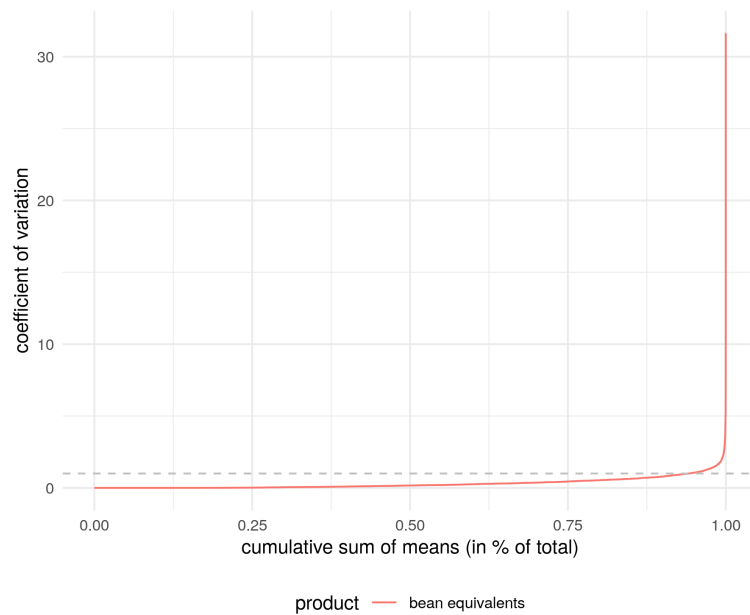


Figure A6: Variability of flows from origin municipalities to importing countries compared to the cumulative sum of flows ordered by CV. The dotted horizontal line corresponds to a value of 1.

or long-distance flows). In addition, more substance could be added by also including the road quality weighting method based on speed limits in the sensitivity analysis, and by considering correlations between the mode-specific cost parameters (i.e., the fact that road transport tends to be more costly than rail and water transport across all sources). Such in-depth examinations are, however, outside the scope of this thesis and are thus left to potential future refinements.

B Benchmarking appendix

B.1 Summary of soy exports by destination

Table B1: Total soy export volumes by destination in kilotons across models in 2013. "trase" gives the total exports of soy produced in Brazil in soybean equivalents according to TRASE. "trase known" excludes exports with unknown origin. "euclid" and "multimode" give the modeled exports of soy produced in Brazil in soybean equivalents according to the Euclidean and multi-modal model and can differ due to different usages of re-exports. "comex" gives total exports (including re-exports) in soybean equivalents according to the COMEX database. "bean", "oil", and "cake" give total COMEX exports (including re-exports) of each soy product separately.

		trase	trase known	euclid	multimode	comex	bean	oil	cake
CHN	China	32782.5	29253.1	32739.7	32737.8	32827.2	32251.8	536.2	25.9
BRA	Brazil	23922.6	23922.6	24031.3	24022.8	24143.7	3673.0	5702.3	14299.1
NLD	Netherlands	5916.8	5344.8	5930.9	5932.6	5943.2	1585.9	10.0	4247.4
ESP	Spain	2214.7	2173.2	2209.9	2209.9	2212.6	1962.7	0.2	244.0
THA	Thailand	2060.4	1647.7	2004.1	2004.6	2008.1	1063.2	0.0	923.2
FRA	France	1749.2	1346.8	1722.4	1723.0	1737.8	149.7	6.1	1545.6
DEU	Germany	1612.7	1602.5	1588.2	1591.8	1600.5	317.9	10.1	1243.1
KOR	Republic of Korea	1585.3	1499.3	1499.8	1500.3	1508.9	350.5	0.2	1131.7
VNM	Viet Nam	1027.0	950.8	1021.4	1021.8	1023.8	571.1	5.4	436.9
TWN	Taiwan	989.0	921.9	984.2	984.1	987.4	979.8	0.0	7.5
JPN	Japan	827.9	799.8	845.5	845.4	845.7	610.6	0.0	229.7
IRN	Iran	765.8	723.2	754.5	755.3	766.5	131.3	85.1	535.5
ITA	Italy	729.3	631.8	722.8	723.7	726.7	356.1	0.0	362.1
GBR	United Kingdom	640.6	540.9	637.5	638.0	639.2	455.3	0.0	179.7
IDN	Indonesia	608.4	580.9	602.4	602.7	604.0	0.5	0.0	589.7
SVN	Slovenia	506.1	460.2	497.0	497.5	500.6	0.3	0.0	488.9
SAU	Saudi Arabia	398.4	307.0	398.4	398.4	398.4	398.4	0.0	0.0
NOR	Norway	404.9	390.6	329.4	329.4	329.4	315.2	0.0	13.9
USA	United States of America	324.6	294.6	324.0	324.0	324.6	324.5	0.0	0.1
MEX	Mexico	286.2	275.9	286.2	286.2	286.2	280.9	5.1	0.0
ROU	Romania	253.0	228.6	250.3	250.4	251.7	73.3	0.0	174.4
IND	India	249.9	206.5	248.7	248.6	251.4	0.5	245.2	0.0
PRT	Portugal	242.2	229.8	241.0	241.0	241.9	201.5	0.0	39.4
ARE	United Arab Emirates	120.5	93.2	204.0	204.0	204.9	26.8	0.0	174.1
TUR	Turkey	178.6	153.6	178.3	178.3	179.3	133.5	0.0	44.7
DNK	Denmark	164.5	135.4	162.5	162.7	163.3	0.0	0.0	159.6
CUB	Cuba	73.9	73.9	137.5	137.7	140.2	0.0	75.4	61.6
DZA	Algeria	133.5	102.6	133.8	133.8	134.1	0.0	114.5	16.5
IRL	Ireland	121.2	119.3	120.3	120.3	120.3	0.0	0.0	117.5
BGD	Bangladesh	113.4	97.1	112.9	112.9	113.8	40.6	62.7	8.7
HRV	Croatia	99.2	96.1	98.2	98.2	98.5	0.0	0.0	96.2
MYS	Malaysia	90.9	26.5	93.3	93.3	93.3	93.3	0.0	0.0
EGY	Egypt	78.1	75.6	77.8	77.8	78.3	0.0	59.5	17.0
POL	Poland	77.9	77.9	77.2	77.2	77.4	0.0	0.0	75.6
ECU	Ecuador	65.6	39.8	64.9	64.9	65.2	0.0	0.0	63.7
MAR	Morocco	50.3	45.1	51.2	51.1	51.6	0.0	50.5	0.0
TUN	Tunisia	39.9	30.1	39.6	39.7	39.9	0.0	17.0	21.9
FIN	Finland	39.5	39.5	39.5	39.5	39.5	39.4	0.0	0.1
VEN	Venezuela	30.8	30.8	38.1	38.1	38.1	20.8	16.9	0.0
COL	Colombia	27.1	27.1	29.3	29.3	29.6	0.0	2.6	26.3
SGP	Singapore	24.2	2.0	24.2	24.2	24.2	22.7	1.5	0.0
ROW	ROW	16.0	16.0	16.0	16.0	16.0	6.2	0.0	9.5
ISR	Israel	15.1	8.1	15.1	15.1	15.1	15.1	0.0	0.0
HKG	Hong Kong	12.5	8.7	12.5	12.5	12.5	8.7	3.8	0.0
URY	Uruguay	3.0	3.0	11.9	11.9	11.9	0.1	11.4	0.2
GRC	Greece	10.8	10.8	10.7	10.7	10.7	0.0	0.0	10.5
BOL	Bolivia	0.0	0.0	10.3	10.3	10.6	0.0	10.3	0.0
PER	Peru	0.0	0.0	10.1	10.1	10.1	0.0	9.8	0.0
AUS	Australia	6.1	6.1	9.7	9.7	9.7	0.0	5.7	3.8
SEN	Senegal	8.0	8.0	8.0	8.0	8.0	0.0	7.8	0.0

Table B1: Export volumes by destination in kilotons across models in 2013 - continued

PRY	Paraguay	0.0	0.0	7.5	7.5	7.9	2.3	5.5	0.0
DOM	Dominican Republic	6.2	6.2	6.2	6.2	6.2	0.0	6.1	0.0
PHL	Philippines	6.1	4.4	6.0	6.0	6.0	0.1	0.0	5.8
ZAF	South Africa	4.4	4.4	4.3	4.3	4.4	0.0	4.3	0.0
NZL	New Zealand	0.0	0.0	3.8	3.8	3.8	0.0	3.7	0.0
CHE	Switzerland	3.8	3.8	3.8	3.8	3.8	3.8	0.0	0.0
PAN	Panama	2.1	2.1	2.3	2.3	2.3	0.0	2.3	0.0
TTO	Trinidad and Tobago	0.0	0.0	1.8	1.8	1.8	0.0	1.8	0.0
SUR	Suriname	0.0	0.0	1.4	1.4	1.4	0.0	1.4	0.0
GUY	Guyana	0.9	0.0	1.2	1.2	1.2	0.9	0.3	0.0
AGO	Angola	0.0	0.0	1.1	1.1	1.1	0.0	1.0	0.0
ARG	Argentina	1.0	0.2	1.0	1.0	1.0	0.8	0.0	0.2
CIV	Côte d'Ivoire	0.0	0.0	0.7	0.7	0.7	0.0	0.7	0.0
CAN	Canada	0.4	0.4	0.4	0.4	0.4	0.0	0.0	0.4
CMR	Cameroon	0.3	0.3	0.3	0.3	0.3	0.0	0.0	0.3
LKA	Sri Lanka	0.3	0.3	0.3	0.3	0.3	0.0	0.0	0.3
JOR	Jordan	0.2	0.2	0.2	0.2	0.2	0.0	0.0	0.2
ATG	Antigua and Barbuda	0.0	0.0	0.2	0.2	0.2	0.0	0.2	0.0
BLZ	Belize	0.1	0.0	0.1	0.1	0.1	0.0	0.0	0.1
COG	Congo	0.1	0.1	0.1	0.1	0.1	0.0	0.0	0.1
GHA	Ghana	0.0	0.0	0.1	0.1	0.1	0.0	0.1	0.0
CHL	Chile	0.1	0.0	0.1	0.1	0.1	0.0	0.0	0.1
MOZ	Mozambique	0.0	0.0	0.1	0.1	0.1	0.1	0.0	0.0
DMA	Dominica	0.0	0.0	0.1	0.1	0.1	0.0	0.1	0.0
NIC	Nicaragua	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
VCT	Saint Vincent and the Grenadines	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
GTM	Guatemala	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
HTI	Haiti	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
COD	Democratic Republic of the Congo	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
KNA	Saint Kitts and Nevis	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
ANT	Netherlands Antilles	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
NGA	Nigeria	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
CPV	Cabo Verde	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
PRI	Puerto Rico	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
LCA	Saint Lucia	0.2	0.2	0.0	0.0	0.0	0.0	0.0	0.0
Total	Total	81724.5	75681.5	81699.8	81699.8	81999.6	46469.5	7082.83	27632.6

B.2 Full benchmarking correlation results

Table B2: Correlation between different model approaches and TRASE for all destination countries. Values indicate Pearson correlation coefficients between the respective model's flows and TRASE, with significance according to p-value (≤ 0.001 : ***, ≤ 0.01 : **, ≤ 0.05 : *). Countries with too few observations are excluded. For country abbreviations see table B1.

	direct correlation			focal mean (queen) correlation			focal mean (100km) correlation		
	multimode	euclidean	downscale	multimode	euclidean	downscale	multimode	euclidean	downscale
CHN	0.8201***	0.8203***	0.8156***	0.9045***	0.9071***	0.8941***	0.9238***	0.9269***	0.8828***
BRA	0.3691***	0.3539***	0.4139***	0.6506***	0.6724***	0.6991***	0.5853***	0.5904***	0.6354***
NLD	0.6806***	0.6087***	0.6515***	0.8227***	0.7779***	0.7719***	0.8286***	0.8014***	0.7901***
ESP	0.793***	0.8383***	0.3208***	0.8867***	0.9118***	0.3483***	0.9386***	0.9452***	0.3883***
THA	0.6375***	0.5995***	0.5773***	0.7696***	0.7777***	0.7322***	0.81***	0.7963***	0.7229***
DEU	0.6284***	0.5467***	0.2758***	0.6446***	0.5941***	0.2656***	0.6106***	0.5642***	0.2462***
KOR	0.656***	0.5983***	0.2441***	0.7502***	0.7161***	0.2649***	0.7471***	0.6857***	0.3483***
FRA	0.533***	0.4369***	0.433***	0.6733***	0.6094***	0.5541***	0.6919***	0.649***	0.6089***
VNM	0.7162***	0.6776***	0.5438***	0.8431***	0.8184***	0.7224***	0.8607***	0.8764***	0.6962***
TWN	0.6119***	0.5879***	0.4503***	0.7347***	0.7374***	0.5229***	0.7813***	0.7966***	0.5635***
JPN	0.8831***	0.8756***	0.5849***	0.9394***	0.9282***	0.7478***	0.9424***	0.9292***	0.7146***

Table B2: Correlation between different model approaches and TRASE for all destination countries - continued

IRN	0.5592***	0.4111***	0.3781***	0.7269***	0.4978***	0.4198***	0.6771***	0.5278***	0.427***
ITA	0.8288***	0.8328***	0.5944***	0.8774***	0.8866***	0.7166***	0.9376***	0.9495***	0.6905***
IDN	0.3253***	0.1794***	0.4305***	0.6249***	0.484***	0.5929***	0.608***	0.3797***	0.593***
GBR	0.6567***	0.5879***	0.542***	0.7825***	0.7555***	0.6568***	0.8613***	0.8035***	0.6996***
SVN	0.3893***	0.3374***	0.2948***	0.4813***	0.4419***	0.3668***	0.5699***	0.5744***	0.4461***
NOR	0.9684***	0.9649***	0.3416***	0.966***	0.9634***	0.3891***	0.9933***	0.9933***	0.4251***
SAU	0.7582***	0.6233***	0.2313***	0.85***	0.7861***	0.2885***	0.8584***	0.7393***	0.3456***
USA	0.8135***	0.7746***	0.4005***	0.9172***	0.8897***	0.4473***	0.9596***	0.9555***	0.5242***
MEX	0.8738***	0.876***	0.3671***	0.9596***	0.9392***	0.408***	0.9739***	0.9639***	0.5036***
PRT	0.8562***	0.8137***	0.5248***	0.8905***	0.8373***	0.6013***	0.9757***	0.9657***	0.6543***
ROU	0.4457***	0.3254***	0.3091***	0.5907***	0.5106***	0.3885***	0.8237***	0.799***	0.4118***
IND	0.579***	0.424***	0.3255***	0.6795***	0.5047***	0.4043***	0.73***	0.6325***	0.4657***
TUR	0.9207***	0.9077***	0.3117***	0.9462***	0.9415***	0.373***	0.9841***	0.9803***	0.3411***
DNK	0.6875***	0.6809***	0.3604***	0.8793***	0.8589***	0.6373***	0.7186***	0.6697***	0.6229***
IRL	0.3202***	0.5695***	0.3762***	0.4703***	0.6839***	0.4203***	0.3548***	0.6644***	0.4643***
DZA	0.6295***	0.559***	0.4168***	0.8132***	0.772***	0.6234***	0.7994***	0.6803***	0.6548***
BGD	0.5396***	0.4699***	0.2518***	0.6728***	0.6513***	0.347***	0.7718***	0.6531***	0.5234***
HRV	0.5862***	0.5655***	0.0797***	0.5414***	0.5409***	0.1793***	0.2934***	0.3014***	0.1023***
ARE	0.1396***	0.1255***	0.1647***	0.4003***	0.3365***	0.2249***	0.4916***	0.3923***	0.3784***
POL	0.8733***	0.7693***	0.0547*	0.8539***	0.6987***	0.0123	0.8617***	0.6241***	-0.0296
EGY	0.4696***	0.3917***	0.3062***	0.6489***	0.6001***	0.4013***	0.651***	0.4919***	0.5684***
CUB	0.0624**	0.2601***	0.1784***	0.1283***	0.271***	0.156***	0.0645**	0.2548***	0.1101***
MAR	0.5542***	0.4976***	0.1845***	0.63***	0.5224***	0.3553***	0.8064***	0.7684***	0.5123***
ECU	0.175***	0.0662**	0.1985***	0.1944***	0.1344***	0.2013***	0.1256***	0.1333***	0.1497***
FIN	0.9549***	0.9553***	0.0811***	0.957***	0.958***	0.1433***	0.9754***	0.9753***	0.1119***
VEN	0.2898***	0.2525***	0.0677**	0.5499***	0.612***	0.0865***	0.6743***	0.7508***	0.0321
TUN	0.2487***	0.0905***	0.3508***	0.5353***	0.3873***	0.5491***	0.5615***	0.2979***	0.5671***
COL	0.2846***	0.1678***	0.1696***	0.341***	0.3286***	0.1753***	0.386***	0.3953***	0.1128***
MYS	0.3305***	0.2867***	0.0377.	0.3148***	0.2643***	0.0958***	0.1973***	0.185***	0.1044***
ROW	0.813***	0.7434***	0.1121***	0.8486***	0.7671***	0.1198***	0.913***	0.844***	0.0963***
GRC	0.5144***	0.064**	0.0174	0.6386***	0.1977***	0.0032	0.7828***	0.3533***	-0.0025
HKG	0.6716***	0.6605***	0.3258***	0.6843***	0.6709***	0.4513***	0.8137***	0.7786***	0.5127***
ISR	0.3906***	0.3383***	0.0756***	0.5185***	0.3945***	0.0845***	0.5604***	0.4347***	0.0371.
SEN	0.4435***	0.6355***	0.2553***	0.7258***	0.9026***	0.3893***	0.5593***	0.7472***	0.3494***
DOM	0.2288***	0.0393.	0.1505***	0.2401***	0.1166***	0.1669***	0.1103***	0.1025***	0.1232***
AUS	0.3181***	0.3092***	0.0349	0.4317***	0.4409***	0.0499*	0.4671***	0.5122***	0.04.
PHL	0.1525***	0.0997***	0.0441.	0.2558***	0.1866***	0.0738**	0.3396***	0.2933***	0.0459*
ZAF	0.643***	0.4925***	0.0677**	0.7685***	0.3461***	0.1017***	0.7945***	0.323***	0.0503*
CHE	0.7651***	0.7336***	0.1728***	0.7799***	0.7495***	0.1372***	0.6802***	0.6403***	0.1205***
URY	0.0784***	0.0707**	0.0423.	0.1521***	0.157***	0.0418.	0.1843***	0.2107***	0.0207
PAN	-0.0042	-0.0025	0.0764***	-0.0013	-0.0054	0.0971***	-0.006	-0.0062	0.1011***
SGP	0.0451*	0.0055	0.0888***	0.1021***	0.0312	0.1432***	0.0813***	-0.0046	0.1125***
CAN	0.1932***	0.1991***	0.017	0.3564***	0.3732***	0.0309	0.6008***	0.655***	0.0131
CMR	0.9117***	0.924***	0.175***	0.977***	0.9721***	0.1813***	0.9441***	0.9125***	0.1394***
LKA	0.0209	-0.0029	0.0467*	0.0814***	-0.0137	0.0632**	0.3237***	0.0565*	0.0585**
JOR	0.0067	0.3437***	0.1679***	-0.0088	0.4088***	0.1291***	-0.0066	0.2937***	0.0952***
ARG	-0.0028	-0.0011	-0.012	-0.006	-0.0062	-0.037	-0.0068	-0.0274	-0.0895***
COG	0.912***	0.9244***	0.175***	0.9777***	0.9735***	0.1813***	0.9464***	0.9159***	0.1394***
NIC	0.1932***	0.1991***	0.017	0.3564***	0.3732***	0.0309	0.6008***	0.655***	0.0131
GTM	0.1932***	0.1991***	0.017	0.3564***	0.3732***	0.0309	0.6008***	0.655***	0.0131
CHL	-0.003	-0.002	-0.0083	-0.0131	-0.0085	-0.0279	-0.0405.	0.047*	-0.0719**
BOL	-0.0016	-0.0015	-0.0083	-0.0069	-0.006	-0.0279	-0.0287	-0.0242	-0.0719**
Total	0.7073***	0.6919***	0.6944***	0.8628***	0.8637***	0.8525***	0.859***	0.855***	0.8238***

B.3 Full benchmarking RMSE results

Table B3: RMSE between different model approaches and TRASE for all destination countries. Values indicate the RMSE between the respective model's flows and TRASE. For country abbreviations see table B1.

	direct RMSE			focal mean (queen) RMSE			focal mean (100km) RMSE		
	multimode	euclidean	downscale	multimode	euclidean	downscale	multimode	euclidean	downscale
CHN	32.903	32.794	33.3225	16.1975	15.7368	16.2773	13.511	12.2944	14.7031
BRA	39.6536	41.1469	37.564	19.4137	19.3222	18.3933	18.1054	18.7947	18.5134
NLD	10.538	11.721	10.6785	5.2388	5.796	5.0763	5.0939	5.6725	5.0668

Table B3: RMSE between different model approaches and TRASE for all destination countries - continued

ESP	4.902	4.3835	7.62	2.1727	1.919	4.4145	1.6521	1.5781	4.3777
THA	3.8084	4.1881	3.9581	1.9326	2.053	1.8437	1.5765	1.8793	1.6798
FRA	3.8048	4.7644	3.8614	1.9499	2.3421	1.9507	1.7178	1.8446	1.7249
DEU	7.4937	7.8725	8.9292	4.1982	4.415	5.2338	3.1625	3.3043	3.9008
KOR	2.9783	3.2184	3.9558	1.335	1.4484	2.1767	0.9503	1.0451	1.6077
VNM	2.4956	2.6346	3.062	1.1185	1.1922	1.4893	0.9701	0.9093	1.4027
TWN	1.6726	1.7187	1.9166	0.8241	0.8215	1.0904	0.5058	0.4855	0.8085
JPN	1.4515	1.4906	2.5986	0.6487	0.6881	1.2691	0.6792	0.7223	1.4734
IRN	2.1785	2.5024	2.4254	1.0424	1.3836	1.3829	0.6713	0.8649	0.8443
ITA	1.3714	1.3755	1.9854	0.7956	0.7633	1.0885	0.6296	0.5823	1.3218
GBR	1.3492	1.6404	1.4631	0.7269	0.8228	0.7833	0.5871	0.777	0.6475
IDN	3.1563	3.4955	2.823	1.3342	1.5975	1.3551	1.6036	1.9943	1.6912
SVN	1.1677	1.2649	1.2183	0.6379	0.6842	0.6995	0.4328	0.4386	0.5161
SAU	0.9312	1.2392	1.3736	0.5008	0.6099	0.9017	0.4425	0.6396	0.805
NOR	0.9688	1.0202	3.7513	0.4944	0.5127	1.7995	0.4131	0.4142	3.4478
USA	0.6369	0.7018	0.9926	0.2635	0.3021	0.5736	0.2236	0.2327	0.6567
MEX	0.8965	0.8892	1.7403	0.3258	0.3982	1.0727	0.2959	0.3444	1.1807
ROU	0.7524	0.8575	0.7881	0.3757	0.4238	0.4157	0.1934	0.203	0.3169
IND	0.5684	0.6739	0.6634	0.2765	0.3483	0.355	0.1603	0.1954	0.2354
PRT	0.3999	0.4573	0.6601	0.2071	0.2545	0.3634	0.1141	0.1352	0.4092
ARE	0.7912	0.8546	0.4446	0.3581	0.395	0.2523	0.1618	0.2121	0.1929
TUR	0.3795	0.4064	0.9199	0.1668	0.1695	0.4642	0.1314	0.1416	0.6761
DNK	0.8632	0.8724	1.1208	0.2903	0.3129	0.494	0.5275	0.5647	0.6733
CUB	0.629	0.6574	0.4284	0.4154	0.34	0.2477	0.3271	0.2654	0.1783
DZA	0.4688	0.5147	0.5522	0.1909	0.2162	0.2574	0.2119	0.2674	0.2864
IRL	1.0657	0.8903	1.0344	0.5464	0.4276	0.5326	0.8832	0.7214	0.8899
BGD	0.3531	0.3746	0.4081	0.1648	0.1691	0.212	0.0799	0.1007	0.1131
HRV	0.4418	0.4598	0.533	0.239	0.2482	0.2551	0.2315	0.2399	0.212
MYS	0.3306	0.3579	0.3408	0.1762	0.2039	0.1692	0.1477	0.1549	0.1425
EGY	0.2729	0.2869	0.2925	0.1196	0.1255	0.1434	0.0895	0.107	0.0947
POL	0.2741	0.3564	0.5583	0.1359	0.1903	0.2704	0.0606	0.0958	0.1408
ECU	0.3199	0.4126	0.3028	0.1707	0.214	0.1509	0.1128	0.1204	0.0846
MAR	0.1765	0.183	0.2093	0.0742	0.0827	0.0908	0.0419	0.0454	0.0619
TUN	0.252	0.2788	0.2165	0.0988	0.1189	0.0903	0.1278	0.1606	0.1335
FIN	0.1984	0.1976	0.6569	0.0851	0.0843	0.2848	0.0735	0.0737	0.3277
VEN	0.2715	0.291	0.2703	0.1263	0.1202	0.1494	0.0542	0.0485	0.0666
COL	0.2102	0.2208	0.2161	0.1008	0.1027	0.1049	0.0436	0.0449	0.0507
SGP	0.0808	0.2467	0.0362	0.0553	0.1262	0.0277	0.0377	0.1509	0.0253
ROW	0.0943	0.1119	0.1607	0.0416	0.0539	0.078	0.0415	0.0576	0.101
ISR	0.0442	0.052	0.0403	0.0246	0.0335	0.0259	0.015	0.0211	0.0181
HKG	0.0528	0.0542	0.0658	0.0264	0.0275	0.0309	0.0156	0.0181	0.0204
URY	0.0351	0.0385	0.0205	0.0234	0.0245	0.0147	0.0165	0.0161	0.0133
GRC	0.0728	0.1028	0.0858	0.0359	0.0557	0.0476	0.0184	0.0292	0.0311
BOL	0.0726	0.0782	0.0148	0.0461	0.0513	0.0123	0.0382	0.045	0.0111
AUS	0.0742	0.0745	0.0751	0.031	0.0311	0.0325	0.0166	0.0163	0.0167
SEN	0.0981	0.0849	0.107	0.0375	0.024	0.0519	0.0612	0.0504	0.0716
DOM	0.0494	0.0572	0.05	0.0261	0.0293	0.0263	0.0247	0.0258	0.0246
PHL	0.0439	0.0512	0.0415	0.0199	0.0235	0.0187	0.0106	0.0129	0.01
ZAF	0.0431	0.0461	0.0525	0.0188	0.0263	0.0275	0.0072	0.0117	0.0121
CHE	0.0596	0.0622	0.0852	0.0324	0.0342	0.0488	0.0277	0.029	0.0372
PAN	0.0303	0.0339	0.0293	0.0167	0.0182	0.0158	0.0164	0.0173	0.016
ARG	0.0067	0.0155	0.0035	0.0037	0.0065	0.0018	0.0021	0.0036	0.0011
CAN	0.0066	0.0065	0.0063	0.0027	0.0026	0.0027	0.0011	0.001	0.0013
CMR	0.0029	0.0028	0.0062	0.0011	0.0012	0.0032	0.0005	0.0006	0.0015
LKA	0.0031	0.0036	0.0029	0.0016	0.0019	0.0015	0.0007	0.001	0.0007
JOR	0.0017	0.0017	0.0013	0.0011	0.0007	0.0007	0.0009	0.0007	0.0004
COG	0.0012	0.0011	0.0025	0.0004	0.0005	0.0013	0.0002	0.0003	0.0006
CHL	0.0003	0.0005	0.0001	0.0002	0.0003	0.0001	0.0001	0.0001	0.0001
NIC	0.0008	0.0007	0.0007	0.0003	0.0003	0.0003	0.0001	0.0001	0.0001
GTM	0.0004	0.0004	0.0004	0.0002	0.0002	0.0002	0.0001	0.0001	0.0001
Total	6.8119	6.9967	6.7704	3.3501	3.3346	3.3481	2.9893	3.0014	3.2243

C Auxiliary figures

C.1 Maps of CBS items across municipalities

Figure C1: Soybean CBS maps

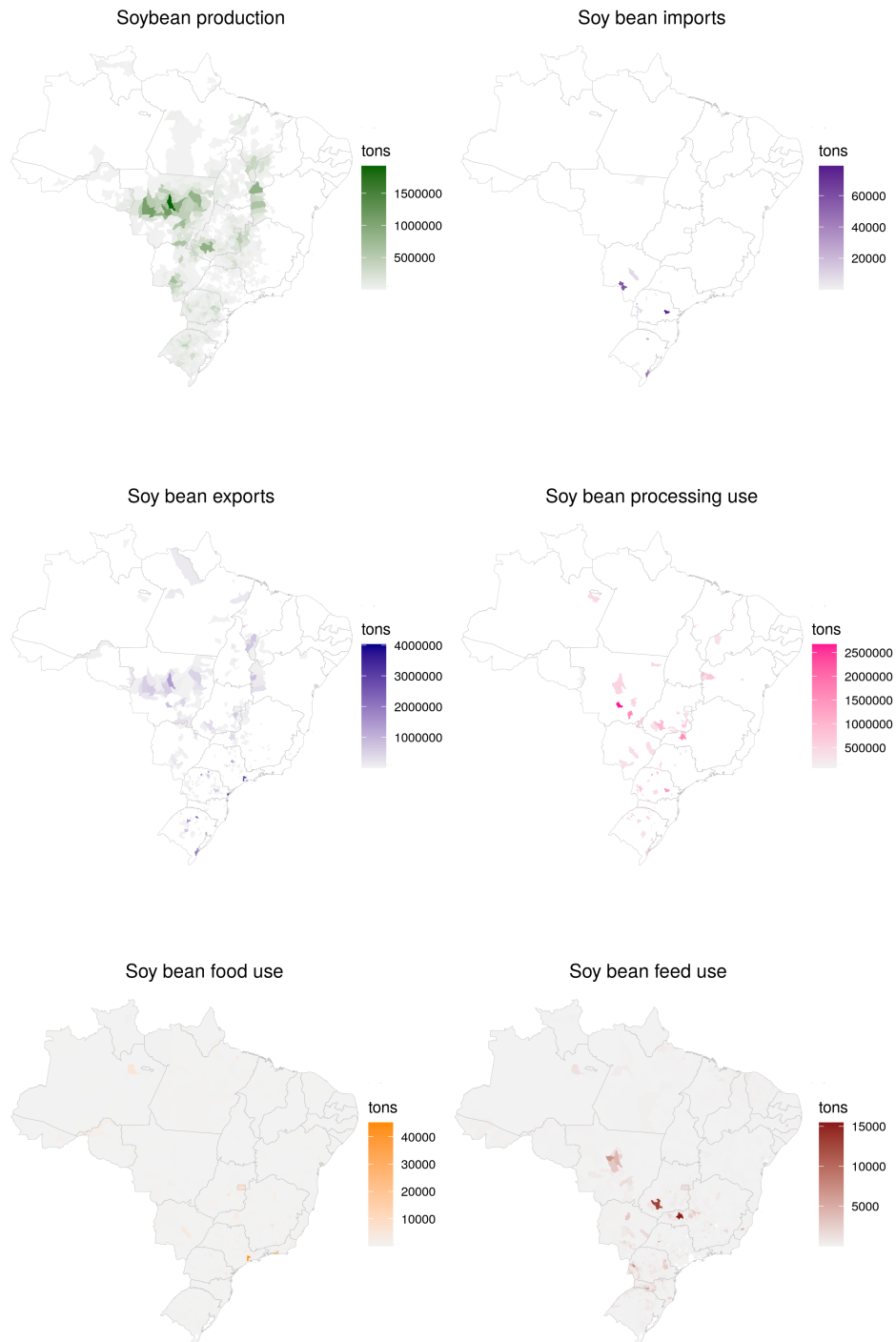


Figure C2: Soybean oil CBS maps

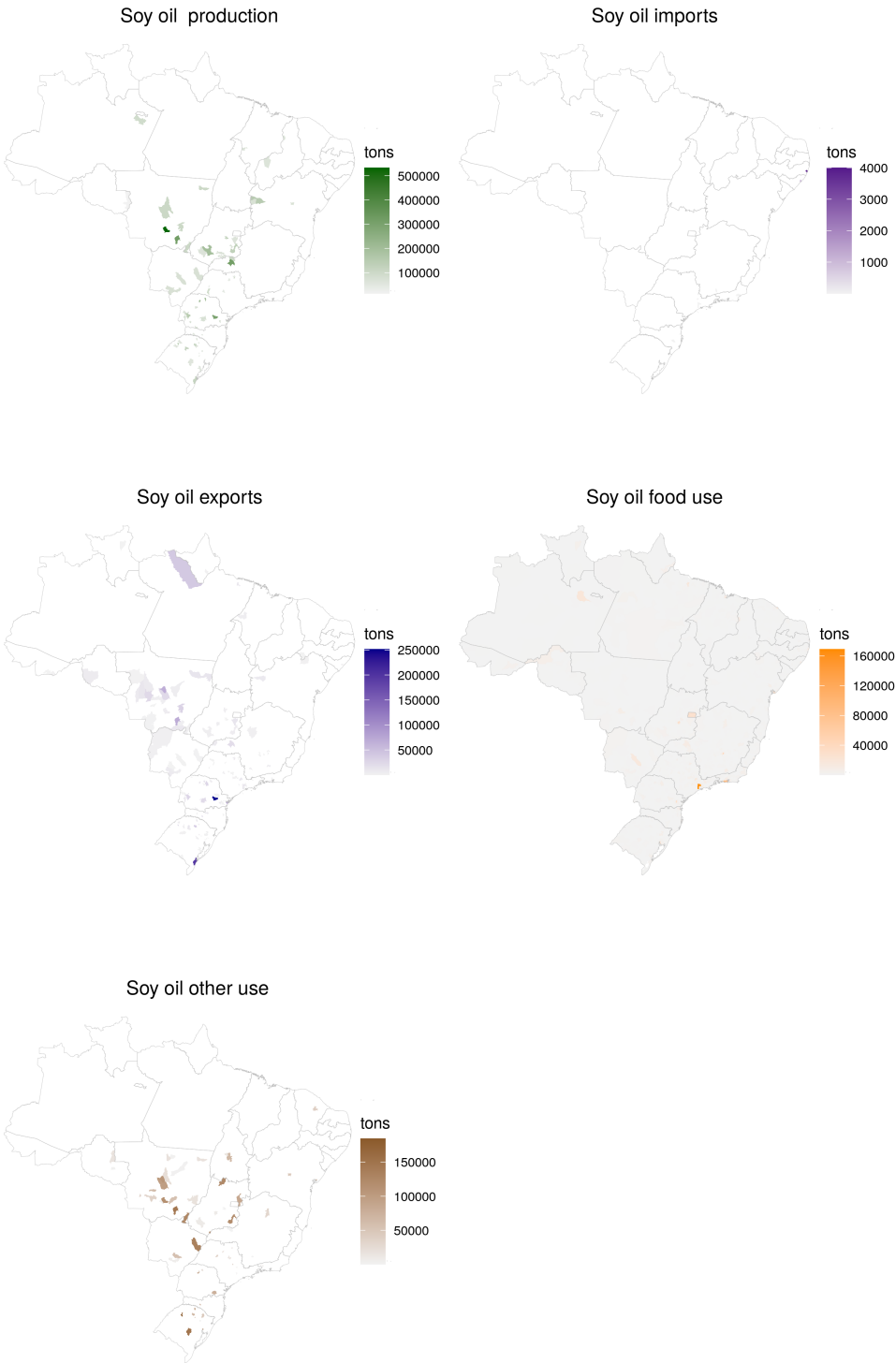
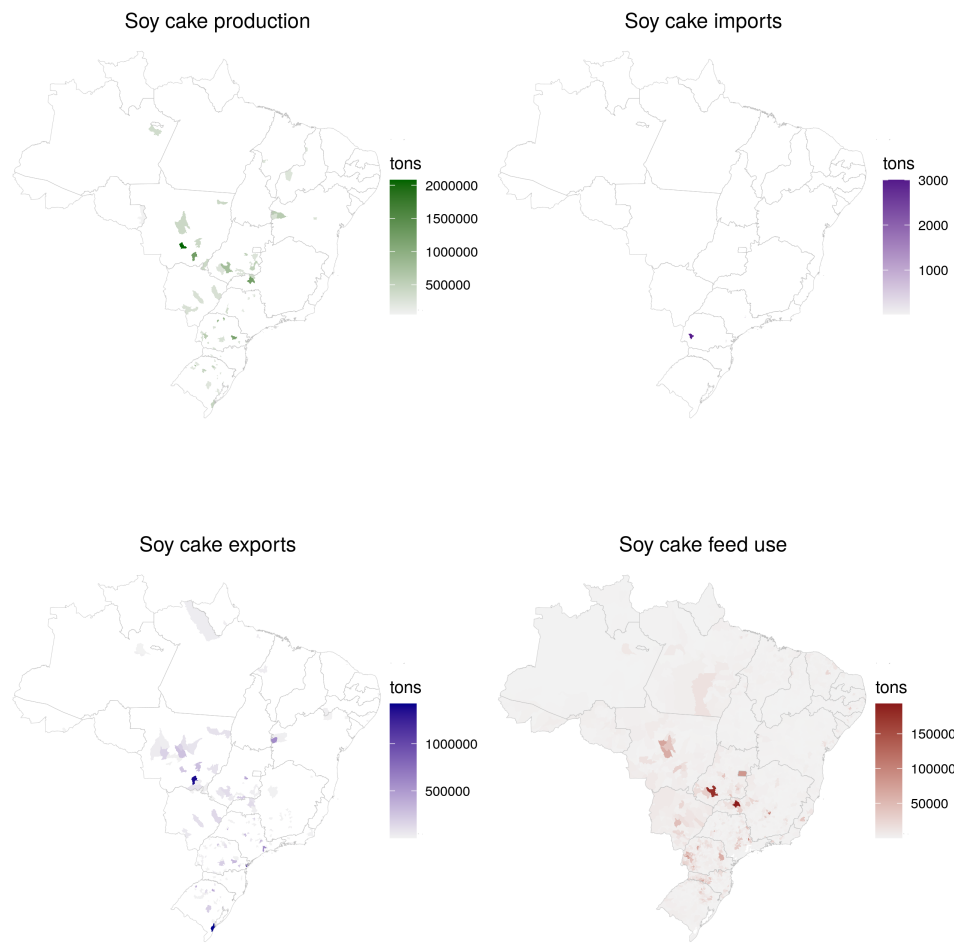


Figure C3: Soybean cake CBS maps



C.2 Benchmarking maps for other main importers

Figure C4: Source of soy consumed domestically in Brazil across the four benchmarks

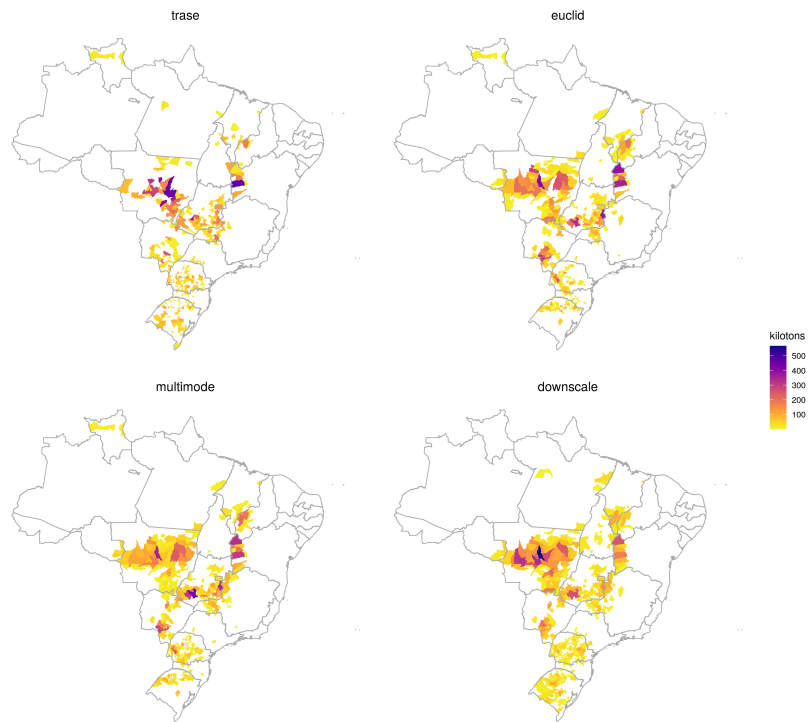


Figure C5: Source of soy exported to the Netherlands across the four benchmarks

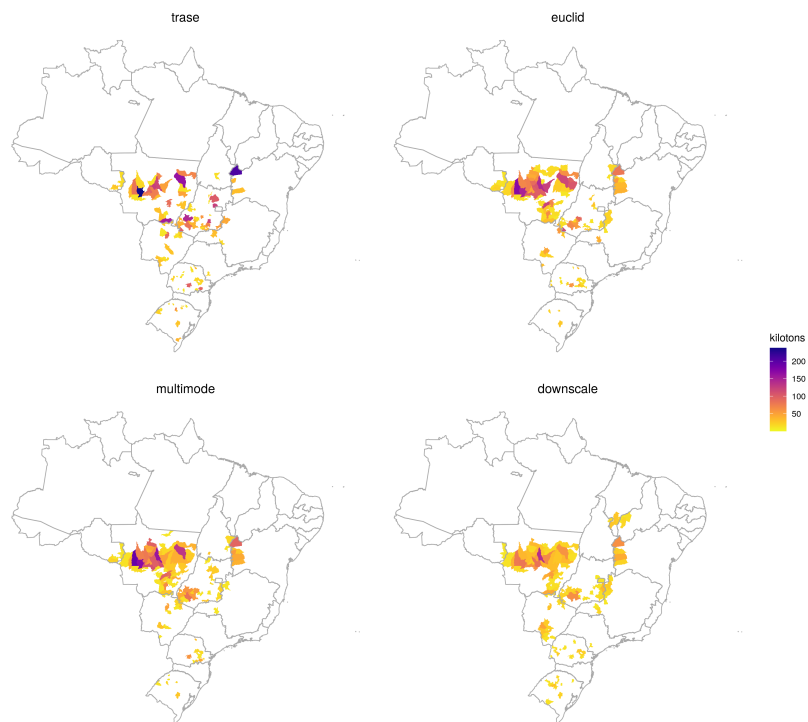


Figure C6: Source of soy exported to Thailand across the four benchmarks

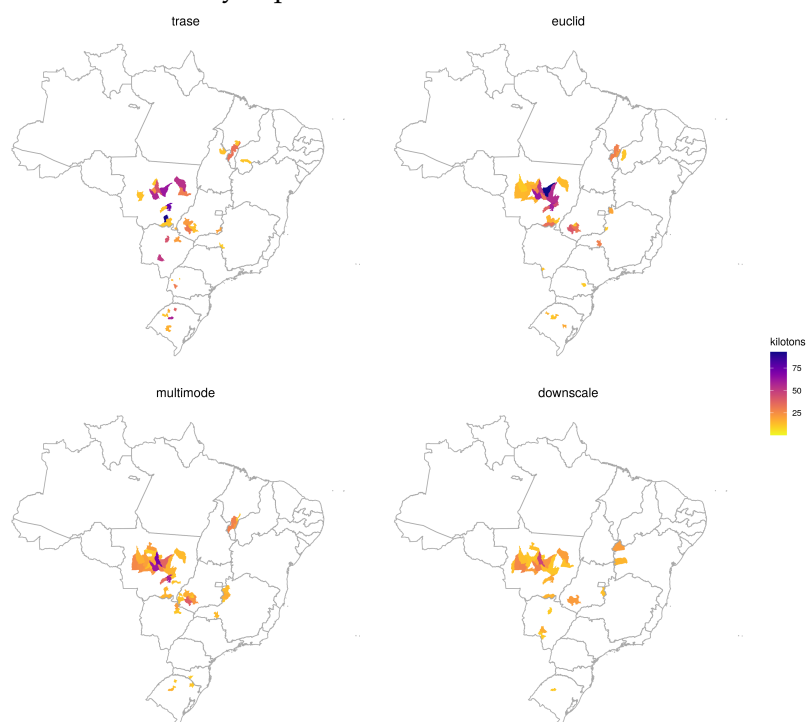


Figure C7: Source of soy exported to Germany across the four benchmarks

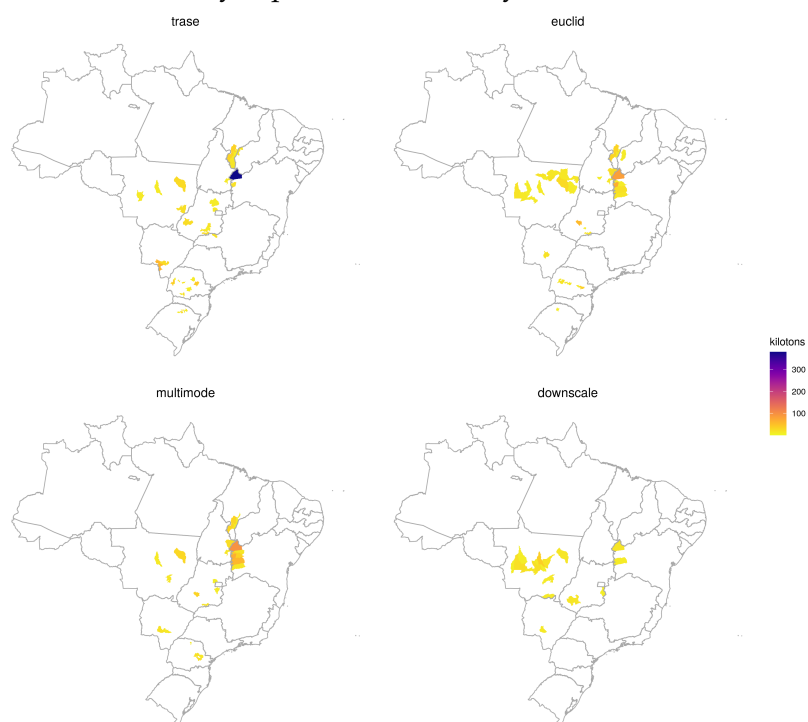


Figure C8: Source of soy exported to Korea across the four benchmarks

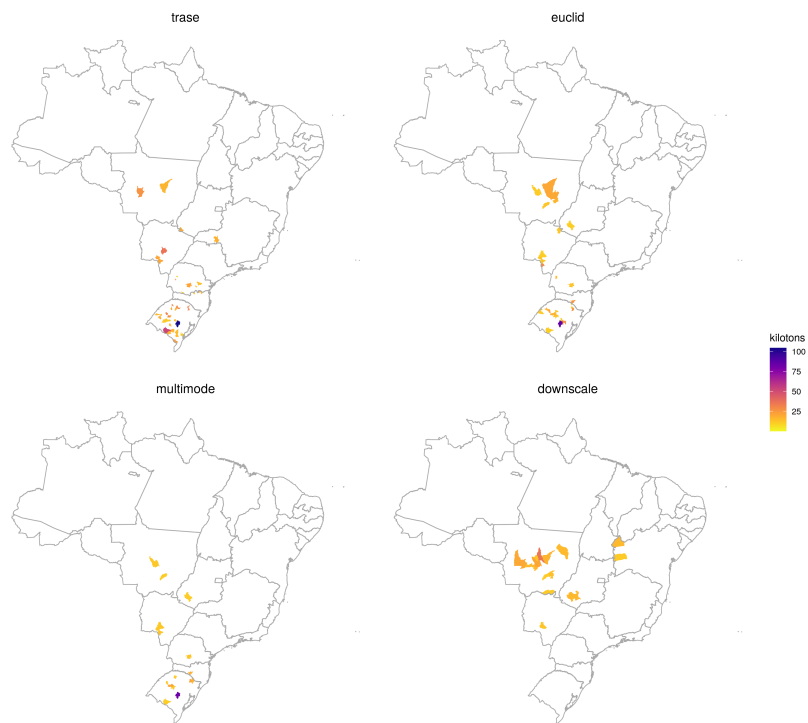


Figure C9: Source of soy exported to Norway across the four benchmarks

